

SQL SERVER DBA

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* Database Environment Life Cycle

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- 1st — Designing phase → DB Designers, DB Architectures.
 2nd — Implementation phase → Scripting Logic Code → DB Developers
 3rd — Maintenance phase → Administering, installation → DB Admin

For Real Time projects

→ Development DB System

→ Testing DB System

→ Production/OLTP System Eg: Banking
Tele Com domain, Hospital, Commercial* Systems to Manage as a DBA

① Development DBA

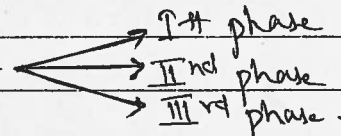
② Production DBA

→ OLTP Systems

→ OLAP Systems / DW Systems / BI Systems

③ BI DBA

For Reports, Analysis, Historical Data, etc.,

* Development DBA* Production DBA— IIIrd phase

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* Roles & Responsibilities of Production DBA

Roles of Production DBA

1. Installation & Configuration
2. Security → Server level, DB level → Encryption, Decryption, Logins.
3. Maintenance & Automation Module
4. Standby Servers Setup [For High Availability]

* log shipping

* Mirroring

* Replication

5. Monitoring & Troubleshooting

* Connectivity issues

* Performance issues.

* Roles of BI - DBA

MSBI Administration

SSIS

SSRS

SSAS

— installation

— Configuration

— deployment

— Security

— Monitoring

* Clustering Practical Implementation

* OS level

classmate * Server level

PAGE



* DBA Related.

7. Service Oriented Architecture (S.O.A)

- Service Broker (SSSB) → Asynchronous Messaging
- Notification Service (SNS)
- DB Mail integration (SMTP)
- HTTP End points.

8. Oracle Integration

9. Database Snapshots → image of a db
→ used for reporting purposes; Read only

10. Data Partitions

11. Database Mirroring → Automatic failover — Redirected to Secondary server

12. Security Enhancements [Keys, Certificates]

13. Maximum no. of instances on single Computer in 2005 — 50 instances.

14. Full Support for 32 bit / 64 bit platforms.

15. Log Shipping & clustering Enhancements

16. Online Indexing.

17. Peer to Peer Replication → used for Network Load Balancing.

Note : 1 - 6 For Developers purpose
7 - 17 For DBA purpose

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2. Work Station Component

- * SSMS - SQL Server Management Studio.
- * Configuration Tools $\longleftrightarrow$
 - SCM - SQL Server Configuration Manager
 - SAC - SQL Server Surface Area Configuration
- * Performance Tools $\longleftrightarrow$
 - SQL Profiler
 - DTA $\rightarrow$ Database Engine Tuning Advisor
- * BIDS Tools
- * Connectivity Components
- * Documentation [Books online]

2. Editions of SQL Server.

SQL Server 2005/2008 is available in 6 editions and can be differentiated by considering the following list

* Features Set

* Price

* O/S Compatibility

* Hardware Minimum Support.

(a) Mobile Edition : * Only for small devices [Mobile device, pocket PC etc,]
* Windows CE, Windows XP Tablet PC

(b) Express Edition : * Free Edition
* Limited DB Engine features only
* No MSBI features
* No Client Tools
* All Windows O/S

(c) Workgroup Edition [Small Organization] $\rightarrow$ 32 bit only
* Limited DB Engine features
* Limited SSIS Development
* Client Tools
* All Windows O/S.
* Price \$ 750

(d) Standard Edition [Medium Organization] $\rightarrow$ 32 bit / 64 bit
* DB Engine features
* DWH/BI Modules (SSIS, SSRS, SSAS)
* No Advanced DB Engine features
- DB Snapshots, Data partitions,
- Online indexing, Advanced SSIS/SSAS/SSRS.
* Only 2 node clustering possible.
* Max. 4 processors — Not Scalable
* Licensing price for one server — \$ 5000

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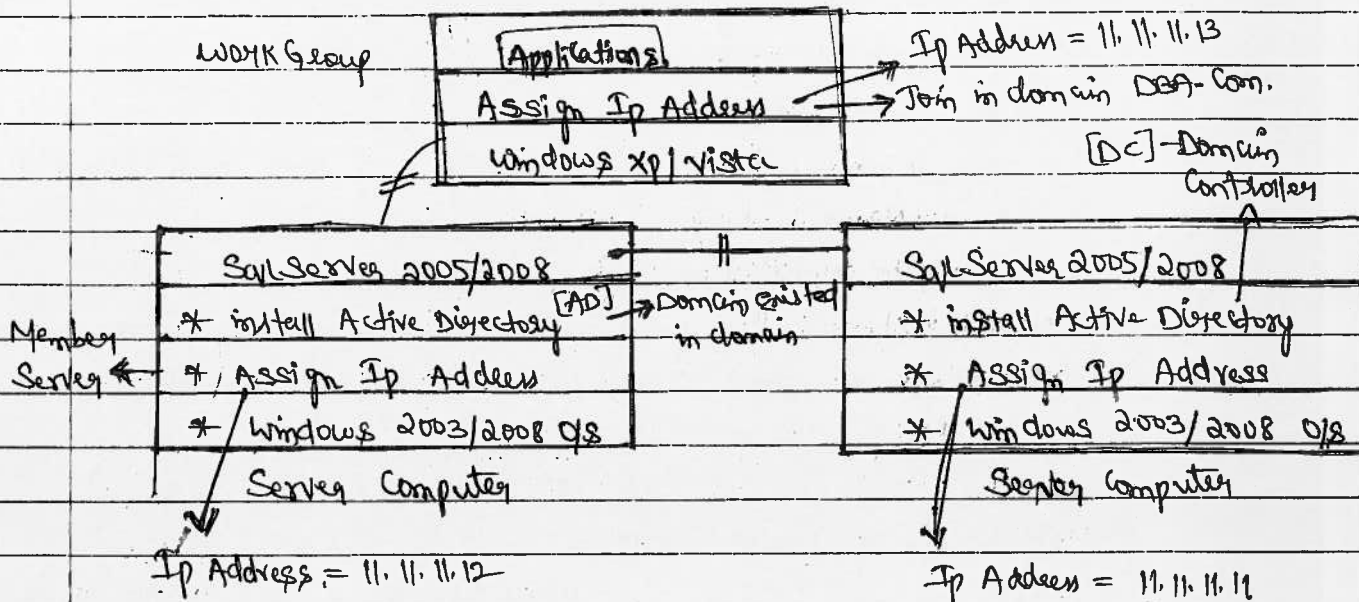
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④

* Installation Process.

5. Setup the Operating System → Server O/S



O/S Level Job

Server Computer: Domain Controller / Member Server

* Install the server O/S of windows 2000/2005/2008.

* Promote Work Group as Domain Controller / Member Server.

* Assign Ip Address

* install Active Directory → Start → Run → dc-promo

Client Computer: Work Group

* install the windows xp/vista O/S.

* Assign the Ip Address

* Join in the existing domain

O/S

On Server Computer

* Install Sql Server all components

(Server components + client Tools)

Sql Server

On Client Computer

* install only client components.

Sql Server

* 1. Practical : Installing the default instance [Server Computer]

* Open SQL Server 2005 S/W

X 86 → 32 bit

X 64 → Itanium (IA) → 64 bit

* Two Sub-folders

① Servers

② Tools

* Goto "Server" folder and double click on "Setup.exe"

* Pre-Requisites installed

(.net F/W, SQL Native client, Setup Support files)

* System Configuration check

* Components to Install screen Appears

/ Check the Required Components

/ SQL Server DB Engine

/ SSIS

/ SSRS, SSAS, SSNS.

/ Workstation Components & Development Tools.

* Click "Advanced" button to change the installation path

* Instance Configuration

— default instance (or)

— named instance

* Specify Service Account

/ Use built-in System Account (or)

/ Domain Account

* Service Account means O/S user Account

Under this account context, Services will be running on the Computer

/ For Stand Alone Computer, Local System Account is Recommended.

/ For N/W, Domain Account is Recommended.

* To Create a Domain Account, click on

/ Start -

/ All programs

/ Administrative Tools

/ Active Directory Users & Computers

To Connect to Named Instance

Server Name = Computer / Instance Name

= Peers / Testing

= Peers / Server 2

Verification: Start → programs → MS SQL Server 2005

→ Verify Client Tools

Open [O/S] Services window

Start → Run → Services.msc

SqlServer MSSQLServer → Default Instance

SqlServer (Testing) → Named Instance

* Practical: Installing client Components / Tools only on client Computer

* Follow the installation process.

* Components List Screen Dialogue box appears.

* Check only Work Station Components & Development Tools.

* Practical: Installing Connectivity Components only on client Computer.

* Follow the installation process.

* Components List Screen Dialogue box appears.

* Check only workstation Components & Development Tools.

* Now click on "Advanced" button

* Cancel the different Components Under 'client Components' except - Connectivity Components.

* Continue the installation process.

* Trouble Shooting Installation Failure:

* Open Installation Summary Report

C:\program files\Microsoft SQL Server\90\Setup Bootstrap\Log

↓

Summary.txt file

↓

SQLSetup XXXX - ComputerName - ComponentName - Log

↓

eg: SQLSetup 0042 - peers - SQL - log

Core

SQL

AS, RS, NS

↓

Analysis Services, Reporting Services, Notification Services

② Configuration and Optimization.

SQL Server 2005 provides 2 Configuration tools.

- (a) SQL Server Configuration Manager (SCM)
- (b) SQL Server Surface Area Configuration (SAC)
- (c) Surface Configuration Manager (SCM)

Open SCM using the following steps

Start → programs → MS SQL Server 2005

Select 'Configuration Tools' and then

Select 'SCM'.

(i) START/STOP/RESTART the Services

Select "SQL Server 2005 Services"

Select particular Instance Service and Right click and then

Select Start/Stop/Restart

Right click → properties (change Service Account)

(ii) Server protocols Configuration

Select "SQL Server 2005 Network Configuration"

Select protocols for a particular instance.

Right click on Protocol → Select Enable/Disable.

Right click on particular instance service changed.

(iii) Client protocols Configuration

Select 'SQL Native Client Configuration'.

Select "Client protocols"

Enable/Disable particular protocol

Four Protocols :

1. Shared Memory : Supports only local connections
2. Named pipes : Supports local & Remote connections.
Low performance & less Security
3. Tcp/Ip : Supports local & Remote connections
High performance & More Security.
4. VIA (Virtual interface Adapter) :
Same as Tcp/Ip
Designed for non-windows OS and
non-tcp/ip networks.

* SQL Server Environment

Database Engine Organizes the databases into 2 Categories:

1. System Databases
2. User Databases

In SQL Server 2000, built-in databases

1. System Databases
Master ; MSdb ; Model ; Tempdb

2. User Databases
PubS ; Northwind.

In SQL Server 2005, built-in databases

1. System Databases
* Master * MSdb * Model, * Tempdb * Resource

2. User Databases
* Empty ; * optionally we can setup
→ Adventureworks
→ AdventureworksDW databases.

* Purpose of System Databases

① Master : Master database is startup database.

it maintains : * Logins information

System Tables	* Databases information
↳ Syslogins	* Configurations
Sysdatabases	* Linked Servers information
Sys Servers	* Permissions, Roles info etc.
Sys processes	
Sys messages	

② MSdb : This maintains * Jobs, Alerts, Operators
* Schedules.

System Tables	* Database mail Configuration
↳ Sysjobs	* Backups History, Jobs History
Sys operators	* Maintenance plan History etc.
Sys alerts	
Sys Schedules	backupset
	backup file
	...
	Sysjobhistory

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Development Tools in SQL-Server 2005

1. SSMS - SQL Server Management Studio → Query Editor
→ Object Explorer → GUI
2. SQLCMD - Command Line Tool → GUI based.

- * 1. SSMS : SSMS provides
- * Query Editor
 - * Object Explorer → GUI based.

SSMS allows us to manage SQL Server & MSBI Services

→ How to use Query Editor?

Open SSMS [Start → Run → sqlwmi] → SQL Workbench

* To open Query Editor

* Click on "New Query" button in the Tool bar.

* A Connection window is displayed

↳ Server Type = Database Engine

↳ Server Name = Peers → default instance

= Peers/Server10 → Named instance

↳ Authentication = Windows Authentication

↳ Click on "Connect" button

Note : To write Comments in T-SQL

(1) - Single Line Comment Eg : -- Sample Notes

(2) /* Multi-Line Comment */

To Execute T-SQL Script → Press F5 (button) function key

To Show or Hide Results windows → press ctrl + R

Examples : * List out Current Server Name, Login information

* How to find Current SQL Server Properties?

Serverproperty(Property Name)



* ServerName * productVersion

* Collation

* Product level → Returns Service packs info

* Edition

* isclustered 0 - non-clustered
1 - clustered.

* Instance Name

* isSingleUser

0 - multi-user mode

1 - single user mode

→ Service packs info

Note: Product level : * RTM No Service pack

* SP1

* SP2

RTM - Release to Manufacture.

Right click on 'New Database' and type

Default path of Database files created

<Home directory>: | MSSQL | Data → datafile → .mdf extension
 (Command Line Tool) → log file → .ldf extension.

In SQL Server 7.0/2000

* OSQL $\rightarrow$ ODBC.

* SQLcmd (new) $\rightarrow$ high performance

- * Used to interact with SQL Server from Command prompt and executes T-SQL scripts at the Command prompt.

(Start $\rightarrow$ Run $\rightarrow$ cmd)

C:\SqlCmd -E -SserverName

(Q2) C:\sqlcmd -UUsername -Ppwd -SserverName.

27. T-SQL Script

Note: 'Go' Represents end of batch

it will submit the T-SQL statements to the SQL Server.

Options : - F → Windows Authentication

- U → SQL Server Authentication

- P → SQL Server Authentication

-S → Server Name

- i → input script file

— 0 — output script file.

Ex : How to use SQL cmd ?

C: /7SayLand -E -Speers/Server10

17 use Master

27 G10

3) Select name, cdate

4) From Sys databases

57 60

Note: To Exit Sql Server

c:\

17 Exist

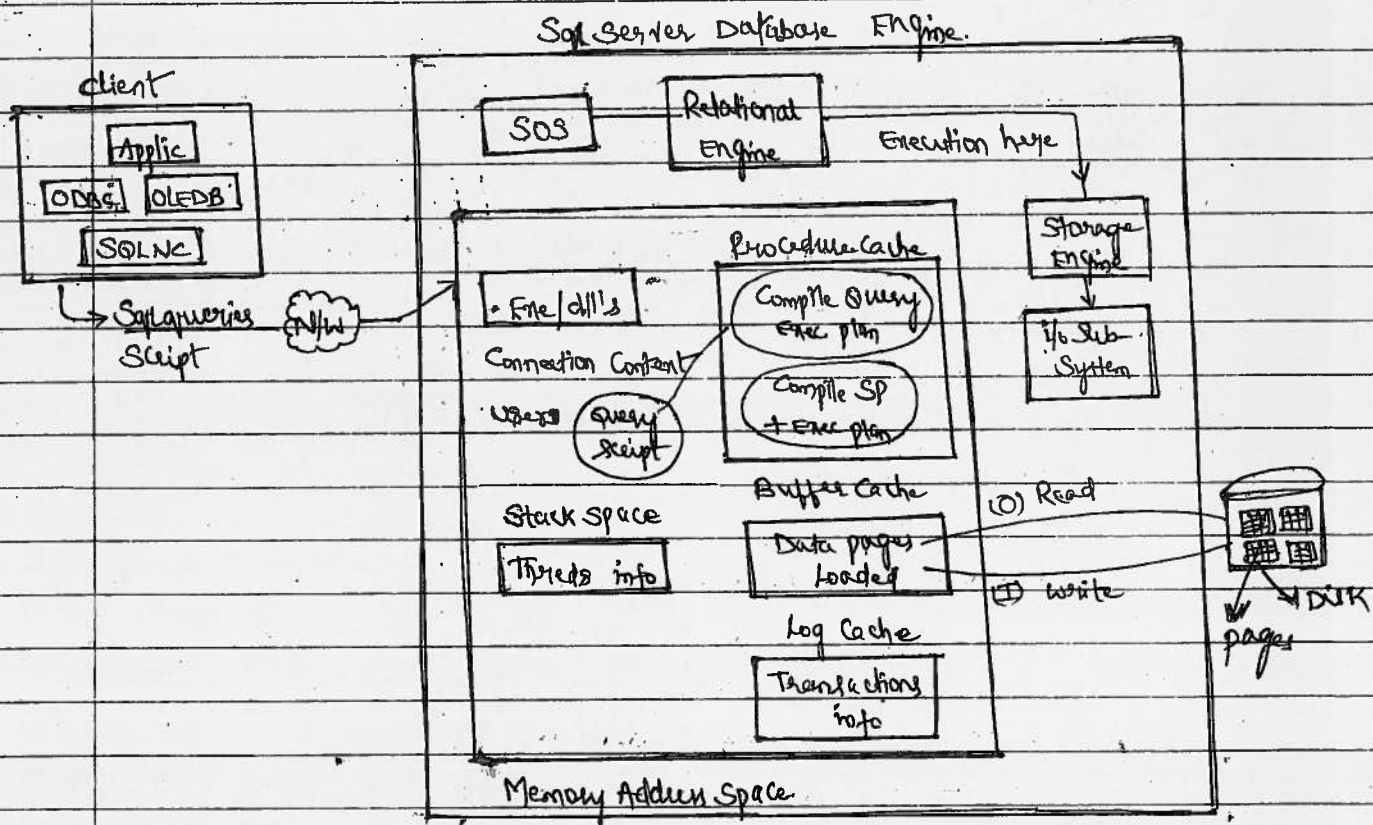
④ Verify the Network Connectivity

* open the Command prompt on Server Computer

* C:\>

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* SQL Server Architecture



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* Minimum Memory per Query — Means how much memory is allocated for single query submitted by the user.

By default 1 MB is allocated.

Threads Configuration [SOS] → SQL OS

* SOS creates a thread to process the user request.

* Each thread takes memory of 512 Kb.

* Minimum no. of worker threads by default — 32, 767.

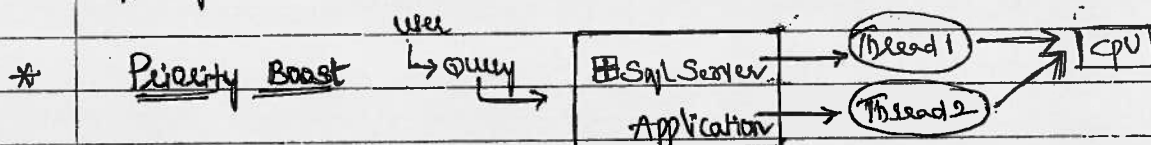
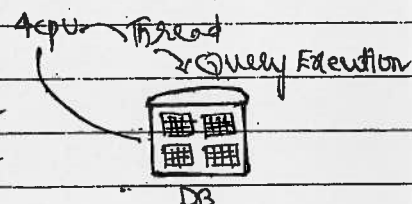
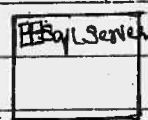
Ex: Concurrent users load on particular SQL Server — 1500

1) Exec Sp-Configure 'Max. worker threads', '1600'

ReConfigure
Go.

Processor's Configuration (CPU):

		0	1	0	1
Affinity Mask — ⑤	P ₁	P ₂	P ₃	P ₄	
Affinity Mask — ⑩	1	0	1	0	



* SQL Server will support by default SMP — Symmetric Multi Processing.

* Affinity Mask decides which processor is allocated for execution purpose.

* Affinity I/O Mask decides which processor is allocated for I/O.

By default, dynamically processors will be allocated for execution & I/O.

* Priority Boost means SQL server threads provided with the highest priority on the server computer.

Ex: $\bar{P}_1 \bar{P}_2 \bar{P}_3 \bar{P}_4$ — 1 0 1 0 — 10 → P₁ P₃ allocate I/O
 0 1 0 1 — 5 → P₂ P₄ allocate I/O

1) Exec Sp-Configure 'Affinity I/O Mask', '10'

ReConfigure
Go

2) Exec Sp-Configure 'Affinity mask', '5'

ReConfigure
Go

3) Exec Sp-Configure 'Priority Boost', '1' → To Enable

ReConfigure
Go.

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* Practical : How to start SQL Server in Single user Mode?

* Open SCM SQL Server Configuration Manager

* Select SQL Server Services

* Select particular instance of SQL Server. (Eg: Server1)

* Right click on properties of ServerName/instance.

Select "Advanced" option

In Startup parameters

..... ; -m and click OK.

Note: -m here changes to multi-user mode.

* Restart the DB Engine service.

* Single user mode means only one user [Sys Admin] can be connected.

* This user can open only one connection. to SQL Server

* [DAC] Dedicated Administrative Connection : → using Command Prompt
→ using Management Studio.

* New feature in SQL Server 2005.

* DAC Connection will use a special thread, separate port no.

* DAC will give faster connection when compared to normal connection.

* Only 1 DAC Connection can be established at a time. i.e. only Sys Admin

Usage : Generally, Administrator will establish DAC to resolve the server or db level problems.

Q. How to Establish DAC?

* We have 2 ways to establish DAC.

(1) SSMS

→ Connection window

→ Server Name = admin : peers / Server10

(2) Sqlcmd

(Windows Authentication)

C:\> Sqlcmd -E -S ServerName -A

→ DAC

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* Properties of a Database File : File Specification

- ① Name = Logical Name. This should be different for each file in particular DB
- ② FileName = physical path of the file.
- ③ Size = initial size of the file
- ④ MaxSize = Maximum size of the data file.
- ⑤ FileGrowth = Autogrowth of the file.

Eg: Name = 'xyz'
 FileName = 'C:\Hello.mdf'

Size = 10Mb

MaxSize = 30Mb

FileGrowth = 3mb

* Practical : Create a Database with 1 data & 1 log file.

Create database Salesdb

ON

(Name = 'xyz',
 FileName = 'C:\Salesdbdata1.mdf',

Size = 3Mb,

MaxSize = 10Mb,

) FileGrowth = 1Mb

Log ON

(Name = 'abc'

FileName = 'C:\Salesdblog1.ldf',

Size = 2mb,

MaxSize = 7mb,

) FileGrowth = 1mb

* Eg: Create a Database with default properties

Create Database bankdbdata

Default properties : <Home Directory>\data

* bankdbdata.mdf

Size = 10mb

MaxSize = Unlimited

FileGrowth = 1Mb

* bankdbdata-log.ldf

Size = 504 kb

MaxSize = 2 B

FileGrowth = 10% Size.

→ path will be in
 SQL Server Home
 direct

-- Ex: Adding a New Log file to a Database.

Alter Database Salesdbdata

Add Log file

(Name = 'log2',

FileName = 'D:\Saleslog2.ldf',

Size = 3mb,

MaxSize = 7Mb,

) FileGrowth = 1Mb

* File Groups * Logical Group of one or more data files in the DB.

* File Groups allows us to organize the datafiles.

In SQL Server, a database can have 2 types of file groups.

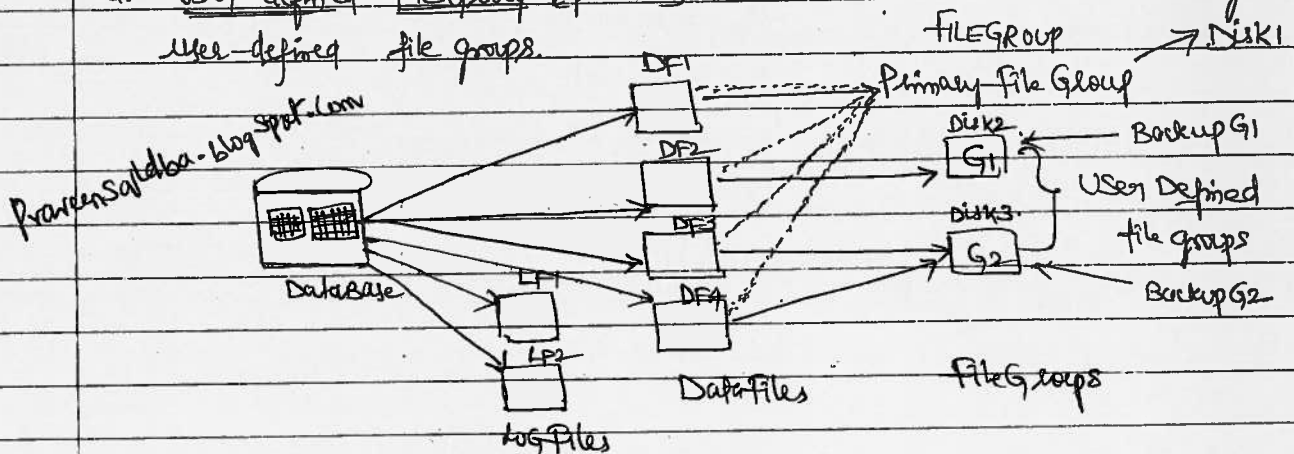
① Primary File Group (Built-in)

② User defined File Group.

File Group Concept can be applied to only data files (only) but not to log files.

1. Primary File Group : A database can have only one primary FG. This is a mandatory in the database.

2. User defined File Group : A database can have many user-defined file groups.



* Advantages of File Groups

1. Backup & Restore of a particular file group will save the time of complete db backup & restore.
2. File Groups allows us to maintain space for the objects (generally for tables, indexes) on specific file group.
3. File Groups can be organized across multiple drives/disks for I/O performance.
4. Data partition implementation requires file groups in the DB.
5. Particular File Group can be marked as Read-only.

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Log ON

```
(
  Name = 'Log1',
  FileName = 'C:\Salesdb\log1.ldf',
  Size = 2mb,
  MaxSize = 10Mb,
  FileGrowth = 1mb
)
```

* Practical 2. Adding File Groups & Data Files to an existing database.
Alter Database [DatabaseName]

Add filegroup fgName

Modify Filegroup fgName
default / read-only.

Remove Filegroup fgName

Add File

(Data File Specification)

To Filegroup fgName.

Eg: Alter Database Salesdb1

Add Filegroup G3

Alter Database Salesdb1

Add File

```
(
  Name = 'S4',
  FileName = 'C:\Salesdb40.ndf',
  Size = 2mb,
  MaxSize = 10mb,
  FileGrowth = 1mb
)
To Filegroup G3
```

* Practical 3. Creating objects (Tables, Indexes) on particular file group.

Create Table TableName

(

Column1 datatype,

Column2 datatype, ...

)

On FilegroupName

Textimage ON filegroupName

* To implement Data partitioning on a New Table we have to follow 3 steps:

Step 1: Identify the partition Column.

Customers → Table Name.

* CustNo — int → partition Column.

* CustName

* Age

Step 2: Prepare partition Function

Partition function decides the range of values to create the partition based on partition scheme.

How to create the partition Function?

Syntax: Create partition Function

AnyName (partitionColumn datatype)

AS RANGE [LEFT/RIGHT]

FOR values (boundary values)

Eg: Create partition Function PFI (int)

AS Range Left

FOR values (1000, 2000, 3000)

Left	Right
$-\infty$ — 1000	$-\infty$ — 999
1001 — 2000	1000 — 1999
2001 — 3000	2000 — 2999
3001 — $+\infty$	3000 — $+\infty$

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Step 3: Prepare the partition Scheme

partition scheme decides physical storage of each partition on a particular file group i.e., mapping each partition with file group

How to create partition Scheme?

Syntax: Create partition Scheme Anyname

AS partition [partition-function-name]

TO [filegroup-name]

Eg: Create partition Scheme PFI

AS partition fgi

TO (G1, G2, G3, G4)

Database Configuration → Properties

① To list all the properties of a database.

Use Master

→ dboption
Exec sp_dboption

② To set the database property

Exec sp_dboption dbname, property, value.

Go

(Or) Alter Database dbname

Set property [ON/OFF]

Practical : Starting & Stopping the Database.

ONLINE : Means start the database

Database is in operational mode.

Offline : Means stop the database.

Database is not in operational mode.

To Start the Database.

Exec sp_dboption 'Salesdb1', 'Offline', 'False'

Exec sp_dboption 'Salesdb1', 'Offline', 'True'

Alter Database Salesdb1

Set ONLINE

Alter Database Salesdb1

Set OFFLINE

Using GUI Mode : Select a particular database

Click on Tasks

Select 'Offline' to stop the database

Select 'Bring Online' to start the database.

User Access Mode for a particular Database.

Eg:

Alter Database dbname

Set Single-user / Multi-user / Restricted-user

↓
only one
'SysAdmin'
user can access

↓
Multiple users
can access

↓
Default mode
when installed.

↓
Multiple 'System Admin'
Users can access.

2. Collation: This property decides

- Language culture Settings
- Sorting order behaviour
- Case sensitivity behaviour, etc.

This can be defined at 3 levels:

① Instance / Server level

only at the time of installation of sql server

② Database level.

* default is inherent from instance.

* we can change it at any time we want

③ Table & Column level

* default: inherent from databases

* we can change it when ever we want.

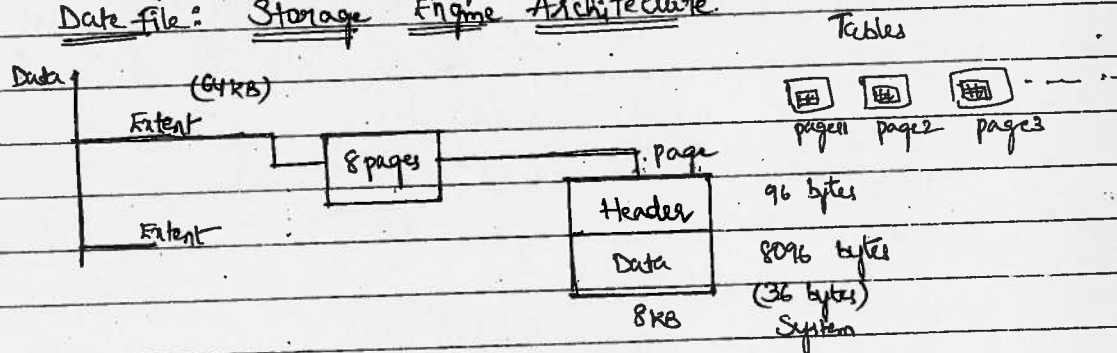
Physical Storage of Database.

Database Means

① Data file (Storage Engine Architecture)

② Log file (Transaction System Architecture)

Data file: Storage Engine Architecture



Extent: This is a logical partition in the data file.

Each extent size is 64 KB.

We have 2 types of Extents:

- ① Uniform Extent: All pages in the extent is allocated for single-object.
- ② Mixed Extent: All pages in the extent is allocated for multiple-objects (more than 1 table or objects).

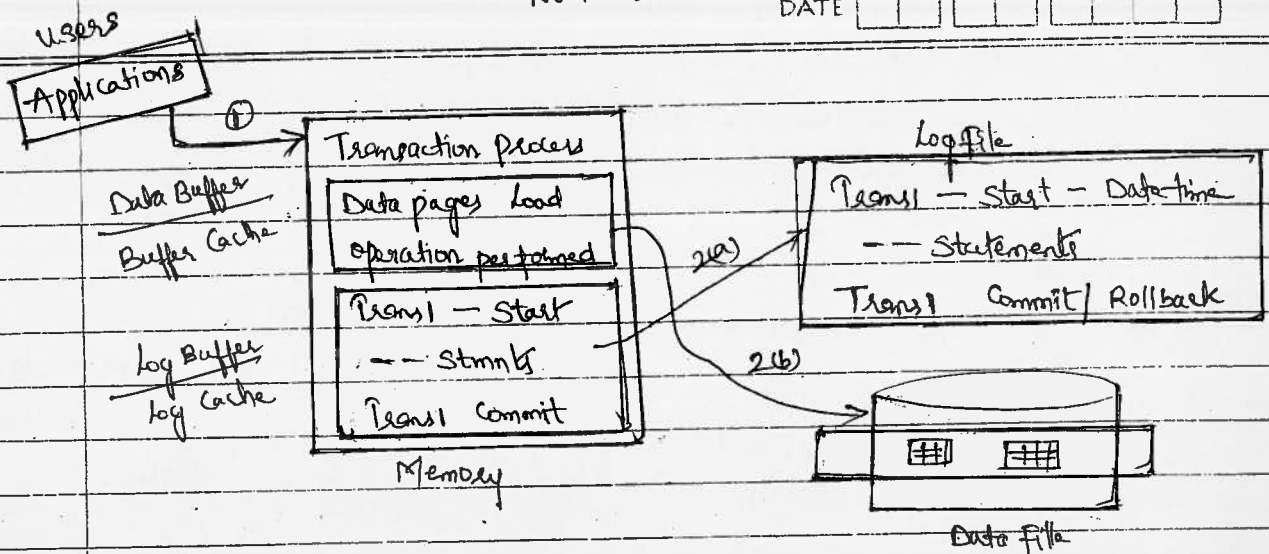
Page: Page means a data block in the Extent.

Each page size is 8 KB.

Page contains 2 Sections.

(a) Header: Address, links to other pages,...

(b) Data: Actual Row Stores.



- Operations performed in the DB tables create transaction process in the memory. Data pages loaded, operations applied & simultaneously transaction details will be stored into log cache.
- Checkpoint signal is generated periodically. Once checkpoint is raised, log buffers transactions are saved into log file on the persistent storage and is called the 'hardening process'.
- Data buffers modified pages are saved into the data file objects for the committed transactions.

checkpoint: Means an internal process periodically executed by the database engine.

Note

- (1) implicit transactions — Auto commit [by default]
- (2) explicit transactions — Begin Trans
Commit

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2. Sql Server Authentication.

In this mode, O/S user must provide separate login & pwd to connect to Sql Server.

Built-in Sql Server login

Username = sa

password = ? (to be provided at the time of installation or after install'n)

Note: windows O/S Administrator's group member, built-in Sql Server login "sa" will have "Sysadmin" role in Sql Server by default.

Practical 1: Setting the pwd / changing the pwd of Sql Server login after installation.

In Object Explorer,

→ Connect to Sql Server with "Sysadmin" privilege.
/ [Windows Authentication] → windows

Practical 2: Changing Authentication mode after installation of Sql Server.

In Object Explorer

→ Connect to Sql Server with "Sys Admin" → (Windows Authn)

Right click on Server and
Select "properties"

In "Security" page

→ Server Auth'n : • Windows Authentication

• Sql Server Auth & Windows Auth

→ Right click on Server and Restart.

→ Mixed mode

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Practical : Sql Server Authentication login — Creation.

* In Object Explorer, Connect to Sql Server with 'SysAdmin' privileges

Expand 'Security' Option.

Right click on 'Logins'

Select 'New Login'

In 'Select a page' dialog box

Select 'General' page option.

Now Select 'Sql Server Authentication'

Login Name = Kiran

password = XXXX

Confirm pwd = XXXX

policy pwd : ☐ Enforce pwd policy

☐ Enforce pwd Expiration

☐ User must change pwd at next login

(win o/s pwd policy is inherited)

Default DB : Master

click OK

Testing : * logon to O/S under any o/s user account.

* Try to Connect to Sql Server under "Sql Server Authentication"

with login Created previously — eg: Kiran

Principal : * This means a login/ application which is accessing Sql Server — resources is called principal.

* windows o/s level [user, Groups]

* Sql Server level [Logins, roles]

* Database level [Users, Roles]

Securables : This means an object in the Sql Server.

They may be — Server level Objects

— Database level Objects

— Schema level Objects

I Server level	II Database level	III Schema level
* Server/ instance	* Database	* Tables
* Logins	* Users	* Views
* Databases	* Certificates	* Stored Procedures
* Endpoint	* Symmetric/ Asymmetric keys	* Synonyms
* Server Roles	* Full-text indexing	* User defined types
	* Schemas	* User-defined Aggregates
	* Service Broker objects	
	* Assemblies	

classmate

Note: Login — Server level

User — Database level.

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(23)

Practical: Mapping login with fixed Server Roles

Eg.

* Login mapping with [Kisan]

[Server Admin
Process Admin
dbcreator]

* In Object Explorer, Connect to sql server with "Sys Admin" privileges.

* Expand 'Security'

* Expand 'Logins'

↳ Select a particular login [Eg: Kisan]

Right click on [Kisan login] it &

Select properties

In [Kisan] properties Dialog box

Under select a page options

Select "Server Roles"

Map the Required Roles

Eg: dbcreator

Server Admin

Process Admin

Public [by default]

Atlast, click OK

2. Database Roles: Database level Securable permissions.

Eg:

login map →

Eg: [Kisan]

Companydb

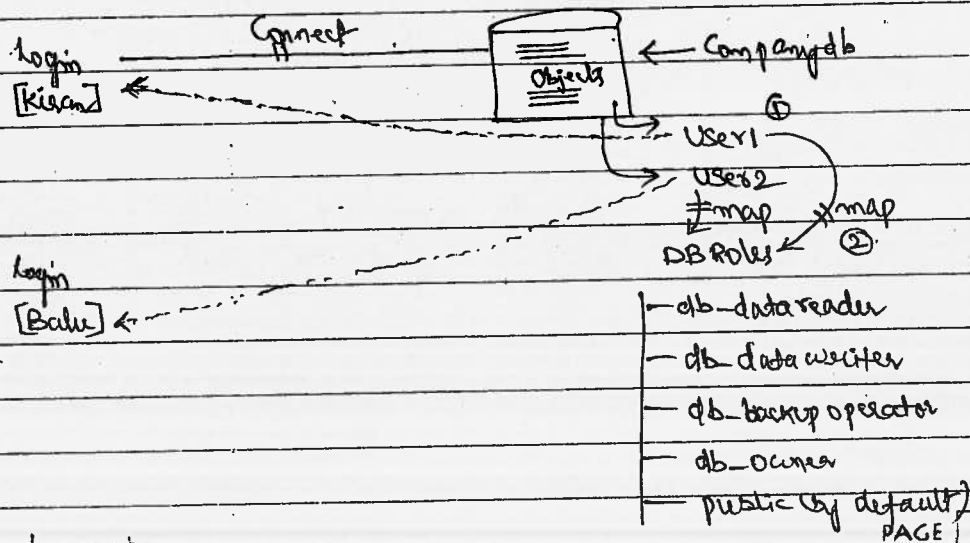
→ database name

☐ dbowner

☐ datareader

☐ backup operator

☐ dbdatawriter



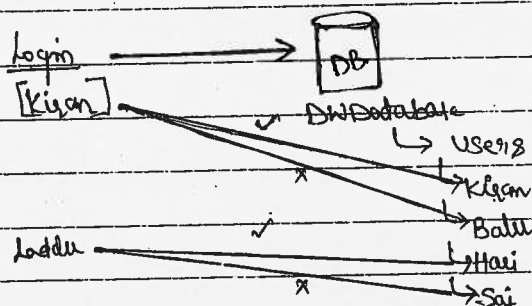
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How to Test: Log on to Sql Server as 'Kiran' and verify the permissions



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Note: In a db, we can create only one db user for a specific login.

3. Custom Roles: Schema level securable permissions

This is also called as 'user-defined' database roles.

In Sql Server, we have 2 types of Custom Roles.

(a) Custom Database Roles.

- * Prepared with a set of permissions on Collection of Objects
- * DB Users can be mapped with Custom 'db roles' directly.

(b) Custom Application Roles

- * Prepared with set of permissions on Collection of Objects.
- * DB Users cannot be mapped with Custom Application Roles directly
- * These roles can be protected with a special password.

Practical: Creating a custom db role and mapping a db user with custom -

* In Object explorer, Connect to Sql Server with Sysadmin privileges role.

* Expand databases

* Select a particular database Eg: DWTDB.

↳ Expand 'Security'

↳ Right click on 'Roles'

↳ Select 'New' → New DB Role

A 'Database Role' dialog box opens.

Select 'General page'

Enter Role Name = abcrole

Owner = dbo

Now Select 'Securables' Page.

Click 'Search' button.

'Add Objects' Dialog box opens.

What

- Specific Objects
- All objects of the type

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↳ Search

↳ Specific Objects

↳ Browse for objects

↳ Select Req. Tables, sp, views

click ok

on the selected objects — Grant the req. permissions.

Click ok.

How a SQL Server Login can access Application Role permissions?

Connect to SQL Server as 'Kiran' for eg.

use dbname

go

Exec sp-setapprole RoleName, password.

Note: Every time a SQL login connected to SQL Server,

sp-setapprole must be executed.

Working with DCL Commands — Data Control Language

These Commands are used to manage the permissions at Schema or object level,

① GRANT

② DENY

③ REVOKE

1. GRANT: This is used to Grant the permission.

Syntax: Grant <Permissions> on Objectname
To dbusername

[with Grant option]

2. DENY: This is used to Cancel the permissions.

Syntax: Deny <Permissions>
on Objectname
To dbusername

3. REVOKE: This is also used to Cancel the granted permissions.

Syntax: Revoke <permissions>
on Objectname
from dbusername

[Cascade]

Revoke is also used to Unblock the denied permission.

with Grant option: This allows the 'user' to grant the given permissions to other user also

-- Connect to SQL server 2005 as an administrator.

USE DWHDB

Go

Grant Select, insert ~~Table Name~~
on dimproductcategory
TO Sqluser1

with Grant option

Testing: Connect to SQL Server as 'SQL User'

USE DWHDB

Go

Result or Access
Tested

Select * from dimproduct category
Go
Grant Select, insert
on dimproduct category
to Sqluser

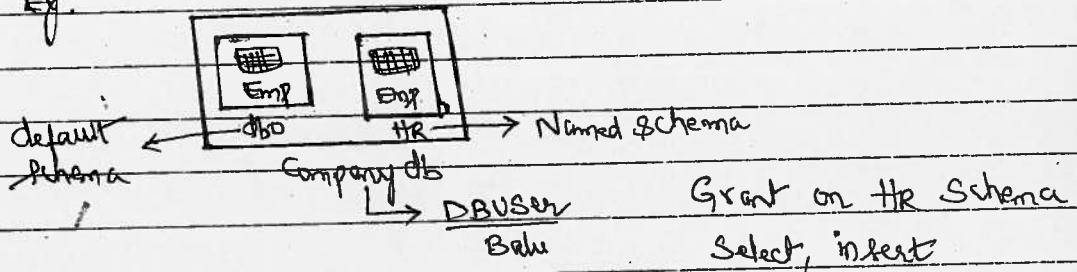
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SCHEMA: This is a logical container in a database used to organize the objects.

- * Schema will act as a 'name space' in the database.
- * Schema contains objects like Tables, Views, Stored procedures, Functions, --
- * In SQL Server 2000, only built-in schema is available.
- * We cannot create named schema.
- * In SQL Server 2005
 - Built-in Schema
 - Named Schema (we can create) → user defined

Eg:



Schema provides 2 Advantages:

1. Avoids naming & conflict problem for database objects.
2. Permissions can be granted at schema level instead of individual object level.

Note: Every Schema should have a owner.

Owner may be a database user in the current database.

- * In SQL Server 2000,
 - Every db user is associated with a schema of 'same name' i.e., 'dbo' was linked with 'Self' schema only.

- * In SQL Server 2005,
 - we have built-in db users/roles as Schemas.
 - ↳ dbo
 - ↳ guest
 - ↳ Information-Schema
 - ↳ sys [New in SQL Server 2005]

* Custom [db] users/roles are separated from Schemas.

* A database user can be mapped with one or more Schemas.

T-Sql Script in Security.

Ex ① Creating Windows Authentication Login in SQL Server. Computer.

Step 1. Create a windows Domain Account in Active Directory users &

Username = praveenm

Password = @peers123

Step 2. Connect to Database Engine with 'Sys Admin' privileges.

Create Login [domain Name/ username]

From windows

with Default-Database = ?,

Default-language = ?

Eg: -- As an Admin

Create Login [dba/Balu] → domainName → Username

From windows

With Default-Database = Master,

Default-language = US-English.

Note: All the properties for windows o/s password policies are inherited into SQL Server.

Ex ② Creating SQL Server Authentication Login.

Connect to DB Engine with Administrator Privileges.

Syntax: Create Login [Login Name]

with password = ?,

Default-Database = ?,

Default-language = ?,

check Expiration = ON/OFF

check policy = OFF/ON.

Eg: Create Login Hasi

with password = '@peers123',

Default-Database = Master,

Default-language = US-English,

check-Expiration = ON,

check-policy = ON

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Full Access of db.

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- Map the above created db user with the db roles required.
 Exec sp-addrolemember 'db_owner', 'haci'.

* ENCRYPTION & DECRYPTION: [Protecting Sensitive Data]

- * Encryption → plaintext converting into ciphertext
- * Decryption → ciphertext converting into plaintext

In SQL Server 2000, Encryption & Decryption

↳ Com dll's — third party tools were used.

In SQL Server 2005, Encryption & Decryption.

↳ Com dll's

↳ Third party tools

↳ .Net Assemblies (CLR integration)

↳ Built-in objects (Keys, certificates)

* Encryption Hierarchy

Windows OS level [DP API]

Server ↓ Server level [Service Master Key]

↳ SQL Server Storage level Encryption.

↓ Database level

↳ Objects (Keys, Certificates)

- * To implement Encryption on db objects level, we have to follow 3 steps.

Step 1: Create the database Master Key.

Syntax: Create Master Key.

Encryption by password = 'Any password'

- * Creating db master key will enable the db to support Encry & Dec

- * A db can have only one Master Key.

Step 2: Create the Key / Certificate

-- Syntax: [Symmetric Key]

Create Symmetric Key [KeyName]

with Algorithm = Any Algorithm

Encryption by password = 'Any password'

Note: We can use Symmetric keys, Asymmetric key, Certificate or — Combination of these.

* DECRYPTION PROCESS

1. Open the Symmetric Key

2. Use 'decryptbyKey' built-in function

decryptbyKey (Column/data)

↳ This returns binary data

↳ Use type conversion functions to convert into target type.

Eg: Step 1: open Symmetric Key Key1

Decryption By

Password = '@Peers123'

Go

Step 2: Select Custno,

built-in function
 ↳ type conversion function.
 Cast (DecryptbyKey (CreditCardno) as int) as CCno
 From Customerdata ← table name

* DIFFERENCE BETWEEN SYMMETRIC & ASYMMETRIC KEY.Symmetric KeyAsymmetric Key.

* Uses single Key for Encryption & Decryption.

* Uses two keys [one Key for Encryption & one for Decryption].

* Based on 'Symmetric Algorithms'

* Based on 'Asymmetric Algorithms'

Eg: DES - Data Encryption Standard

Eg: AES - Advanced Encryption Std.

* CERTIFICATES: A Certificate is a security object which can be used for encryption/decryption.

A Certificate can be used along with Symmetric or Asymmetric Key.

-- Syntax: Create Certificate Certif-name

Encryption By password = 'Any pwd'

with Subject = AnyText

-- Built-in functions: ① EncryptbyCert (parameters)
 ② DecryptbyCert (parameters)

Combination: Creating Symmetric Key using Certificate

Create Symmetric Key [Keyname]

with Algorithm = Any Alg.

Encryption By Certificate Certifname

* An orphan user means a 'db' user not associated with any sql login.

* Orphan users problem is raised in 2 cases.

① Master Database Damaged & we don't have recent backup i.e. logins are lost.

② when we copy database from one server to another server, only 'db' users are copied but logins are not copied.

To list out orphan users in a database, we use the following:

System: Use DBName
Go

EXEC sp_change_users_login 'Report'

To Resolve Orphan user, we follow the following steps:

— **Step 1**: Create a new sql login / Copy the sql login.

Eg: Create Login Haei } : sql server login
with password = '@preet23'

— **Step 2**: Map the orphan user with the sql login just created above.

Eg: Use DBname
Go

EXEC sp_change_users_login 'update-one', 'dbuser', 'loginname' ① ② ③
(db) EXEC sp_change_users_login 'Auto-fir', 'dbuser' ① ②

— 'Auto-fir' means the 'db' user is mapped with the sql login of the same name automatically.

Eg: Use DBname

Go

EXEC sp_change_users_login 'Auto-fir', 'Haei'

EXEC sp_change_users_login 'Report'

System: EXEC sp_change_users_login 'Action', 'dbuser', 'login'

Action → Report
 → update-one
 → Auto-fir

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Practical : Setting / Changing the Recovery model of a DB.

GUI : In Object Explorer

↳ Connect to DB Engine

↳ Expand 'Databases'

↳ Right click on a particular 'db'

Select 'Properties'

In 'DB properties' window

In 'options' page

Recovery Model

- Full
- Bulk Logged
- Simple

T-SQL : Alter database dbname

Set Recovery full / simple / Bulk-logged

Recovery model type

Eg: Alter database dwhdb ← any dbname
Set Recovery full

* RECOVERY MODELS :

1. Simple : No operations are maintained in transaction log.

Advantages : * Operations gives better performance

* Log file contains always free space.

Drawbacks : * No Recovery is possible from transaction log

* We can't take transaction log backup

Recommended Environment : * Development / Testing databases.

* Read-only databases (Reporting Databases) in production.

2. Bulk-logged :

* Simple operations are completely maintained in transaction log.

* Bulk-operations are minimally logged.

Adv : Bulk operations give performance.

* Recovery is possible for simple operations from transaction log

Recommended Environ : Datawarehouse system

or : OLAP Server Database.

3. Full : Simple / bulk operations complete information is maintained in the transaction log.

Adv : Recovery is possible for all operations from transaction log.

Dis-Adv : * operations gives less performance

* Log files use more space.

Recomm. Environ : Real-Time production Databases (OLTP)

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* 'Append' means the latest backup set will be created at new position and existing backup sets will not be erased.

* 'Overwrite' means the latest backup set will erase all the existing backup sets and created at first position.

* TYPES OF BACKUPS

SQL Server natively supports 4 types of Backups.

1. Full
2. Differential
3. Transaction Log.
4. File & File Group.

1. Full Backup : This backs up the Complete database (data & log files) onto back media as it currently stands.

Advantages : Complete database backup is available at a particular point of time.

Dis-Advantage : * More time Consuming process.

* More storage space required, if we choose 'append' option.

Note : For every database, first 'backup' must be 'Full' backup [atleast once]

Recommended Option : . overwrite

2. Differential Backup : (Incremental Backup)

This backups the changes made in the database [data & log files] since the last full backup.

Advantage : Time saved when Compared to Full Backup.

Recommended Option : Append.

1. Practical : Implementing Full Backup

GUI Mode In Object Explorer

↳ Expand 'Databases'

↳ Select a particular 'database'

Eg: TESTDB

D:\Testdb.mdf → data file path

E:\Testdb-log.ldf → log file path

↳ Right click on the required database

Select 'Tasks' → Select 'Backup'

In 'Backup Database' window...

In 'General page'; Database = TESTDB

Backup Type = Full

Backup Set Name = TestdbFull

Destination : @ Disk

Click 'Add'

File Name = D:\Testdb-Bak' → OK PAGE

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Click [OK]

Q. 3. Practical : How to perform Differential Backup of a Database ?

GUI Mode : In Object Explorer

↳ Expand 'Databases'

↳ Right click on a particular Database [Eg: Testdb]

Select 'Tasks' → Select 'Backup'

In 'Backup Database' window

In 'Select a page' list box → Select 'General' option.

Source : Testdb

Database (Salesdb)

Recovery Model : Full

Backup Type : Differential

Backup Set :

Name : Testdb differential

Description : Testdb diff. Backup

Destination :

Backup to



DISK

☐ Tape

Click on button

'Select Backup Destination' window comes.

Destinations on Disk

☐ File Name : D:\Testdb diff. bak

☐ Backup device :

Now select options in select a page list box.

Overwrite media :

☐ Backup to the Existing Media Set

☒ Append to the Existing backup set ← Recommended.

☐ Overwrite all existing backup set

Reliability :

☒ Verify backup when finished.

Click

Click

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Q. Practical: How to Restore the Database from Latest Full Backup?

Eg: Restore Database Testdb
From Disk = 'D:\Testdb.bak'
with Recovery, file=1,
Password = '@peers123'

Note: A Database comes into Online with 'Recovery option' once database backup is restored.

Practical: How to Restore the Database from Latest Full Backup and Latest differential backup?

-- Take Tail log Transaction Backup

{ Backup log Testdb
To Disk = 'C:\Testlog.bak'
with NO-Truncate, NORecovery }

'db' will be in restoring state means 'db' is inaccessible.

-- Now we Restore Full Backup

{ Restore Database Testdb
From Disk = 'D:\Testdb.bak'
with NORecovery, file=1,
Password = '@peers123' }

'db' will be in restoring state with latest full backup.

-- Restoring Differential Backup

{ Restore Database Testdb
From Disk = 'D:\Testdb-diff.bak'
with Recovery, file=2 }

'db' Comes onto online applying latest differential backup.

Q. Practical: How to Perform Transaction Log Backup?

Transaction log Backup — backs up only the transactions in the log file since * the last transaction log backup (or)
* the last differential backup (or)
* the last full backup

(T-log)

Transaction Log Backups: These are performed in the sequential order by using log sequence Number (LSN).

Generally T-log Backups require

- ① To reduce the loss of data/ no loss of data when the 'db' is damaged [upto the last minute, data to be recovered].
- ② To perform point-in-time restoring.
- ③ To truncate T-log file periodically for maintaining free space.

Now Select 'Options' in 'Select a page' list box

Overwrite media: ☐ Backup to the Existing media Set

☒ Append to the Existing Backup Set

☐ Overwrite all Existing Backup Set

Reliability: ☒ Verify backup when finished.

Transaction log:

☒ Truncate the T-Log → (Ordinary)

☐ Backup the Tail of the Log → (Tail Log Backup)

Click [OK]

Q.

How to perform Transaction-Log Backup using T-SQL Code?

T-SQL Code: Syntax: Backup Log dbname

To Disk = '.File path'

with option1, option2,

Options: Truncate only → ordinary T-log backup (default)

No-Truncate → Tail log Transaction backup

Recommended option: Append (noinit)

Eg: Backup Log Testdb

To Disk = 'D:\Testdblog.bak'

with Name = 'Testdblog',

Noinit ← append option

(-n can also be used)

• .trn is default file Extension for T-log backup

*

File & File Group Backups:

* This allows to perform a particular file group data files backup

* This can be operated in 2 modes

(a) Full Mode

(b) Differential Mode.

(a) Full: All data files in the file group are backed up onto the backup media as it currently stands.

(b) Differential: The changes made in the data files of a particular file group since the last file group backup are backed onto the backup media.

Practical : How to perform a file & file group backup of a 'db' using T-SQL Code?

T-SQL Code : Syntax : Backup Database DBName
[File | File Group] = 'Logical Name'
To Disk = 'filepath'
with Option1, Option2, ...

Eg: Backup database Salesdb1
File Group = 'G1'
To Disk = 'D:\Salesdb1.bak' ^{overwrite}
with Name = 'G1backup', ^{init}

Note: file & file group backup backs up the particular filegroup datafiles and corresponding transactions in the log files.

* BACKUP STRATEGY

This means different types of backups or required schedules applied or performed for one or more databases.

Eg: ① Daily - Full Backup at 10.00 pm.

② Sunday → Full Backup
weekdays → Differential Backups.

③ Daily full Backup at 8 Am.
Hourly Transaction log Backup

④ Sunday → Full Backup
weekdays → Differential Backups
Hourly → Transaction log Backup

* RESTORING STRATEGY

The process of recovering the database from the available backup sets is called restoring.

We have 2 types of Restoring.

① Current State Restoring

② Point-in-time Restoring

1. Current State Restoring: This means restoring the 'db' upto the last-minute when the database is damaged.

2. Point-in-time Restoring: This means restoring the 'db' upto a specific date & time [earliest state]

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⑤ put the database in multi user mode

Syntax: Alter database dbname
Set multi-user.

⑥ Practical: How to perform Current State Restoring of a database?

① -- Full Backup at 8:00 Am.

Using T-sql Code?

8 Am
Full backup
Emp [1 row]

Backup database Testdb

Note: This db initially consists of

TO disk = 'D:\Testdb.bak'

'Emp' table with few rows.

with Name = 'Testdb-full', init ← (over write)

② Create Table Customers

8:05 Am
diff backup
Customers [1 row]

-- Perform diff. backup at 8:05 Am.

Backup database Testdb.

TO disk = 'D:\Testdb.bak'

with differential, Name = 'Testdbdiff', NoInit

← append to the existing backup file

③ insert few rows into 'Emp' table

inserting few
rows into Emp
& Customers
tables

insert Emp values (20, 'Kumar')

insert Emp values (30, 'Ravi')

insert Emp values (40, 'Gita')

insert Customers values (200, 'Ramu')

insert Customers values (300, 'hari')

④ -- Performing Transaction Log Backup at 8:10 Am

Emp - 3 rows
Customers - 3 rows

Backup log Testdb

TO disk = 'D:\Testdb.bak' ← append

with Name = 'Testlog', NoInit

at 8:10 Am

Transn Log Backup

2 new rows
inserted into
Emp table

⑤ insert Emp values (50, 'madhu')

insert Emp values (60, 'Sini')

⑥ Now suppose that our db 'Testdb' is Corrupted and not accessible.

Note: To Corrupt the particular database manually,
Goto S.C.M

⑦ Select the particular 'MSSql Service' server.
Right click on it and Select 'Stop'

⑧ Now go to the 'Testdb' database data file file path, here
change the name of the data file or delete the data file manually.

Step ③ Run the Tail Log Transaction Backup just before applying the backup set.

eg: Backup Log Testdb
TO Disk = 'D:\Testdbtaillog.tsn' db will be in
with No-Truncate, NoRecovery ← Restoring state

Step ④ Performing Restoring process. → [Applying Required backup sets]

-- Restore Full Backup

Restore Database Testdb db will be in
From Disk = 'D:\Testdb.bak' restoring mode
with file=1, NoRecovery ←

-- Restore Latest Differential Backup

Restore Database Testdb
From Disk = 'D:\Testdbdiff.bak' db will be in restoring mode
with file=1, NoRecovery ←

-- Restore Transaction Log Backup

Restore Log Testdb
From Disk = 'D:\Testdblog.bak' db will be in restoring mode
with file=1, NoRecovery ←

-- Restore Tail Log Backup

Restore Log Testdb
From Disk = 'D:\Testdbtaillog.tsn'
with Recovery ← Now the db comes into 'online' mode

Note: Hence 'Emp' table contains 6 rows
 customers table contains 2 rows

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Restoring StrategyT-Sql Code

Practical: How to perform 'Point-in-time' Restoring of a 'db' using T-Sql Code?

* Implementing 'Point-in-time' Restoring of a database.

Generally, ① To Eliminate junk/wrong data from an inconsistent database.

② To maintain historical database state on demand.

We perform Point-in-time Restoring.

* Take full Backup of a 'db' (eg: Database: Testdb; Tablename: Emp)

C:\sample.bak — 8:07 Am — 6 Rows in Emp table

* 8:10 Am — Transaction Log Backup — 8 Rows information

* 8:13 Am — Transaction Log Backup — 10 Rows information

* 8:16 Am — Transaction Log Backup — 15 Rows information

* Suppose that some invalid rows has been populated into the 'db' from 8:12 Am onwards.

Backup Strategy

full backup

* — Full Backup at 8:07 Am

{ Backup database Testdb } — Full Backup at 8:07 Am.
To Disk = 'C:\sample.bak'

-- Transaction Log Backup at 8:10 Am

T-log Backup{ Backup Log Testdb } Transaction Log Backup at 8:10 Am
To Disk = 'C:\sample.bak'
with Noinit ← append

-- Transaction Log Backup at 8:13 Am

T-log Backup{ Backup Log Testdb } Transaction Log Backup at 8:13 Am
To Disk = 'C:\sample.bak'
with Noinit ← append

-- Transaction Log Backup at 8:16 Am

T-log Backup{ Backup Log Testdb } Transaction Log Backup at 8:16 Am
To Disk = 'C:\sample.bak'
with Noinit ← append

Verify: Select * From Emp.

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[Continuation].....

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Practical : How to perform 'Point-in-time' Restore of a db using GUI mode ?

-- Restoring a 'db' to a 'point-in-time' using GUI.

In Object Explorer

↳ Expand 'Databases'

↳ Select a particular database: (Eg: Testdb)

↳ Right click on the 'db' and

↳ Select 'Restore Database' option

In 'General' option page; *Destination for Restore.

To Database : Testdb ▾

To a point in Time :

--

 ...

← click on this

'Point-in-Time' Restore window appears

Restore TO :✓ ☒ A specific date & time :Date :

12/27/2010

 ▾Time :

8:12:00 AM

 ⚙

* Source for Restore

✓ ☒ From device

--

 ...

← click on this

'Specify Backup' window appears

click on Add button

'Locate Backup files' window appears

Select the file : C:\sample.bak

click OK button.

In 'Select the Backup sets to Restore'

☒ Restore

--

Now Select 'options' page.

Restore options☒ Overwriting the existing database.Restore state : ✓ ☒ Restore with Recovery

Finally click OK finally.

Verification :

Select * from Emp

Hence the 'Testdb' database is restored to a time of 8:12 am

BACKUPS ONTO NETWORK SHARED PATHS

* Create a folder to share

↳ Computer Name = PTPL

Ip Address = 11.11.11.2

↳ D:\peersbackup ←

Right click on the above folder and

Select 'Sharing & Security'

Share Name = Peersbackup

click 'Permissions' button

click on [Add]

Select or Add 'Administrators Group'

Full Control ☒ Allow

click [OK]

* Now Connect to Sql Server [Eg: peers] as an Admin.

↳ Computer Name = peers

— Backup database Salesdb

To Disk = || PTPL\peersbackup\abc.bak'

↳ Computer name ↳ Shared folder name

with Name = 'Salesdb full', init ← (overwrite)

* How to Access Network Shared folder.

|| ComputerName\SharedFolderName.

Backup Device : This means a logical name to represent physical path of a backup file. ~~Single path~~ ← (single paths)

* 'path' may be local / network shared path

logical Name

* How to Create Backup Device using T-Sql Code?

Use Master

disk → tape

Exec sp_addumpdevice 'devicetype', 'logical Name', 'physical path'.

Eg: Exec sp_addumpdevice 'disk', 'abcd', 'C:\program files\Microsoft Sql Server\mssql.1\mssql Backup\abc.bak'

Scenario →

How to Move the User database files from one physical path to another physical path?

* Connect to a SQL Server [Eg: Peers]

* Create a 'db' [Eg: Testdb]

EXEC sp_helpdb Testdb → To get 'Testdb' db Complete information

Step 1. Stop the particular database (Eg: Testdb) ✗

Syntax: Alter Database Testdb

Set offline -- Press F5.

datafile: C:\
logfile: D:\

Step 2: Now we can cut & paste the database files to the target physical location.

Move the data file Eg: D:\

Move the log file Eg: E:\

Step 3: Modify the New file paths information in the System Catalog

Syntax: Alter Database dbname

Modify file

(Name = ?

) FileName = 'New path'

Eg: ① Alter Database Testdb

② Alter Database Testdb

Step (A) Modify file

(Name = 'Testdb',

FileName = 'D:\Testdb.mdf'

) Go

Step (B) Modify file

(Name = 'Testdb_log',

FileName = 'E:\Testdblog.ldf'

) -- Press F5

Step 4: Select the particular 'db' and make it online

Eg: Alter Database Testdb
Set Online

Note: These New paths will be updated in master db system catalog

To Verify the 'db' New file paths :

EXEC sp_helpdb Testdb

-- Press F5.

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* Suppose, 'Master' db original path files as

data file — 'D:\Master.mdf'

Log file — 'C:\mastlog.ldf' DATE

(44)

* Moving the 'Master' Database files from one physical path to another path.

Note: Initially check the path of Master database files.

Step 1: Modify the 'New' file paths information in the System Catalog.

Alter Database Master

Modify file

(Name = 'Master',

file Name = 'D:\Master.mdf'

) Go

Alter Database Master

Modify file

(Name = 'Mastlog',

file Name = 'E:\mastlog.ldf'.

)

-- Press F5 Now the Modified physical paths of the 'Master' db files will be modified in the System Catalog.

Step 2: Stop the SQL Server Service using S.C.M.

Step 3: Cut & paste the 'Master' db files to target physical location.

/ eg: Move the data file to D:\

Move the log file to E:\

- SCM

Note: Now, we need to modify the 'Startup parameters' of instance using

Step 4: Modify the 'Master' db new file paths information in the 'Startup parameters' using SCM of the required Server.

* open S.C.M

* Select a Particular MSSQL Server Service

* Right click on it and Select 'Properties'

* click 'Advanced'

In "Startup parameters":

change the filepaths as follows:

datafile → -d D:\master.mdf;

errorfile → -e C:\Program Files\Microsoft SQL Server\MSSQL-1\MSSQL\LOG\ErrorLog;

logfile → -l E:\mastlog.ldf

Now, click on button

Step 5: Re-Start the SQL Server Service using SCM

— Finally, Verify the New database file paths of the 'Master' db.
EXEC sp_helpdb Master

DATE

**** Define Backup DataBase (Full) Task :**

• Databases

<Select one or more>

Select this

☒ All databases

☐ All system db's (Master, MSdb, Model)

☐ All user db's (excluding Master, Model, MSdb, tempdb)

☐ These databases

☐ Master

☐ Model

☐ MSdb

☐ SQLdb

☐ Testdb

• Destination :

Backup to

☒ Disk

☐ Tape

☐ Backup databases across one or more files :

If backup files exist

☒ Create a backup file for every database.

☒ Create a sub-directory for each database

Folder : D:\Program Files\Microsoft SQL Server\MSSQL.1\MSSQL\Backup [...]

Backup file Extension

☐ Verify backup integrity

click on button

**** Select plan properties**

Define a Schedule for the Maintenance plan

Schedule : Not Scheduled (on demand)

click button

**** 'New Job Schedule' window appears.**

Name = FullBackupEverySunday.

Scheduled type = Start Automatically when SQL Server Agent starts ☒ Enabled
Start whenever the CPU becomes idle.

☒ Recurring

One-time

classmate

PAGE

-- To Execute 'Maintenance Plan' Manually in object Explorer.

In object Explorer

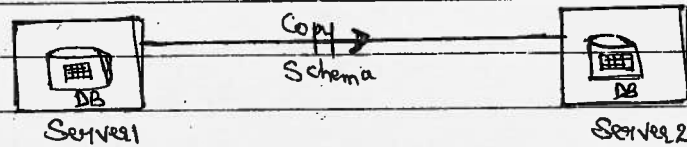
↳ Expand 'Management'

↳ Expand 'Maintenance plans'

↳ Right click on particular/(required) plan Created earlier
Select/Click on 'Execute'

-- Result : Status : Success / Failure information is displayed.

Scenario 4



Scenario-4 Q

How to Copy Database objects & Schema from one server to another server?

① Generate script for database, objects by using 'Generate Script wizard'

② Execute this script file on 'Target Server'.

Step I

(Step 1). Use 'Generate Script wizard'

* Database Script :

In object Explorer

↳ Connect to Required source server (eg: PTPts) ↑ server name

↳ Expand 'Databases'

↳ Select a particular Database (eg: DWHDB)

↳ Right click on the Database - (DWHDB) and

Select 'Script Database' and

Select 'Create to' and

Select 'File'

A 'Select a file' window appears

FileName: D:\DWHDBScript.sql

Click **Save** button.

Stores db & script
↳ (properties of db,
Create db --)

*
*
**

Database objects Script:

In object Explorer

↳ Connect to Required source Server

↳ Right click on 'Management' and

↳ Select 'Generate Script'

A 'Welcome to the Generate SQL Server Scripts wizard' window appears

click **Next** button.

DATE

--	--	--	--	--	--

**** Script Wizard Summary :**

Review your Selections

- Script Wizard Summary.

☒ Server — Peers☒ Database — DWHDB☒ options☒ objects — Scripts all objects☒ Output options — Save to file in Unicode format

↳ D:\DWHDBObjects.sql

Finally, click **Finish****** Generate Script progress**☒ Success

6 Total

0 Error

6 Success

0 Warning

/ click **close** button.Note: Always Exclude the objects with EncryptionStep 2**Step 2:** Execute the Script file above created/saved on the target server.

* Open the 'db' script file (Eg: DWHDB.sql)

* Change the Database file paths to the required location.

↳ (Generally 'Home Directory')

* Open SSMS (SQL Server Management Studio)

* Connect to Target Server in New Query Editor

Here Target: Eg. peers

* Select 'File' menu option.

↳ Select 'open' → Select file.

In 'open file' window

File Name = **DWHDB-Script.sql**Open objects script file in (SSMS) new query and execute itPraveen.madupu@gmail.com
98661-30093.

PraveenSQLdba.blogspot.com

Note: At the time of Restoring a database, we have to change the data file, log file paths to our required locations using the option with MOVE.

Scenario 6

Script Based

Q. How to take a single table backup? (table schema + table data)

In Object Explorer

- ↳ Connect to a particular server eg: Peers
- ↳ Expand 'Databases'
- ↳ Select a particular db (eg: Sales db)
- ↳ Expand this db and select 'Tables' and Right click on a particular required table eg: dbo.dim Customer.
- ↳ Select 'Script [Script-table As]' and Select 'Create To' → file
- ↳ A 'Save As' window opens
- File Name : dimCustomer.sql [in some required path]

Note: Table data is Exported to a flat file.

Table Data Backup Operation Steps

- ① Export table data to a text file
- ② For Import/Export wizard
 - ↳ To open 'SS Import/Export wizard' (SSIS wizard)
 - Start → Run → DTS Wizard

****** choose a DataSource (Select the source from which to copy data)

Data Source : Sql Native client

Server Name : Peers

Authentication

- ✓ Use windows Authentication
- Use Sql Server Authentication.

User Name :

Password :

Database : SWHDB ▼ Refresh

click on Next button.

****** choose a Destination : (Specify where to copy data to:)

Destination : Flat file Destination ▼

File Name : D:\dimCustomer.txt Browse

Locale : ☐ Unicode

Code page :

This file is auto created.

Scenario - 7 : Backup Databases - Additional options

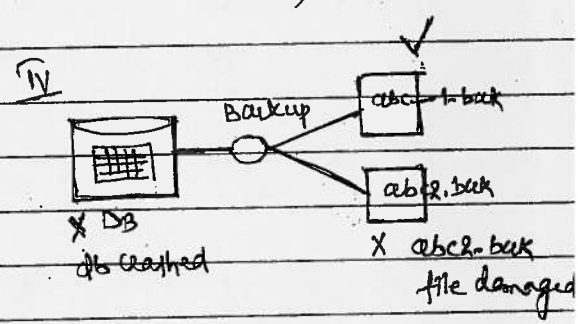
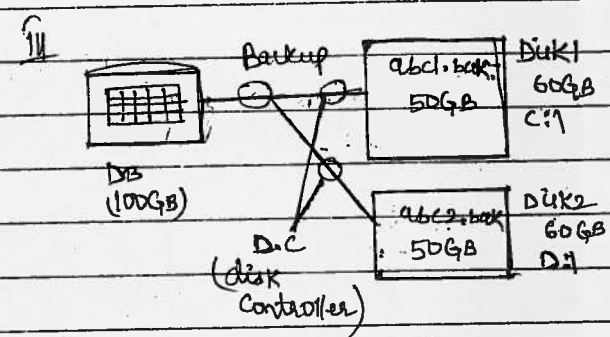
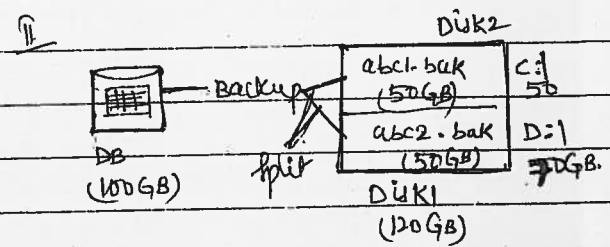
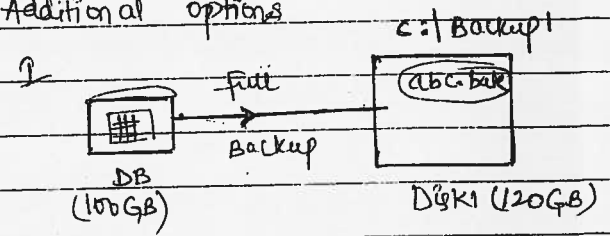
1. Stipped / Split Backup
 2. Mirrored Backup
 3. Copy only Backup
 4. Partial Backup
 5. Encryption & Decryption
 6. Compressing Backups
- *. Multi-File Backup

1. Split / Stipped Backup :
- This backup allows to split the 'db' into multiple files on different drives or different disks.

Advantages : Very faster in backups
Dis-Advantage : To recover the 'db', all the striped backup files must be in consistent state.

Generally, Split file backup is used in 2 cases :

- Case (i) Backup media drives having lack of space compare to 'db' size
- Case (ii) To speedup the backup operation for the large databases.



Q. How to take the Split Backup of a database ?

-- Split back
 Backup database Testdb
 To disk = 'c:\Testdb1.bak',
 'c:\Testdb2.bak'.

-- Restore Split Backup
 Restore Database Testdb
 From Disk = 'c:\Testdb1.bak',
 'c:\Testdb2.bak'

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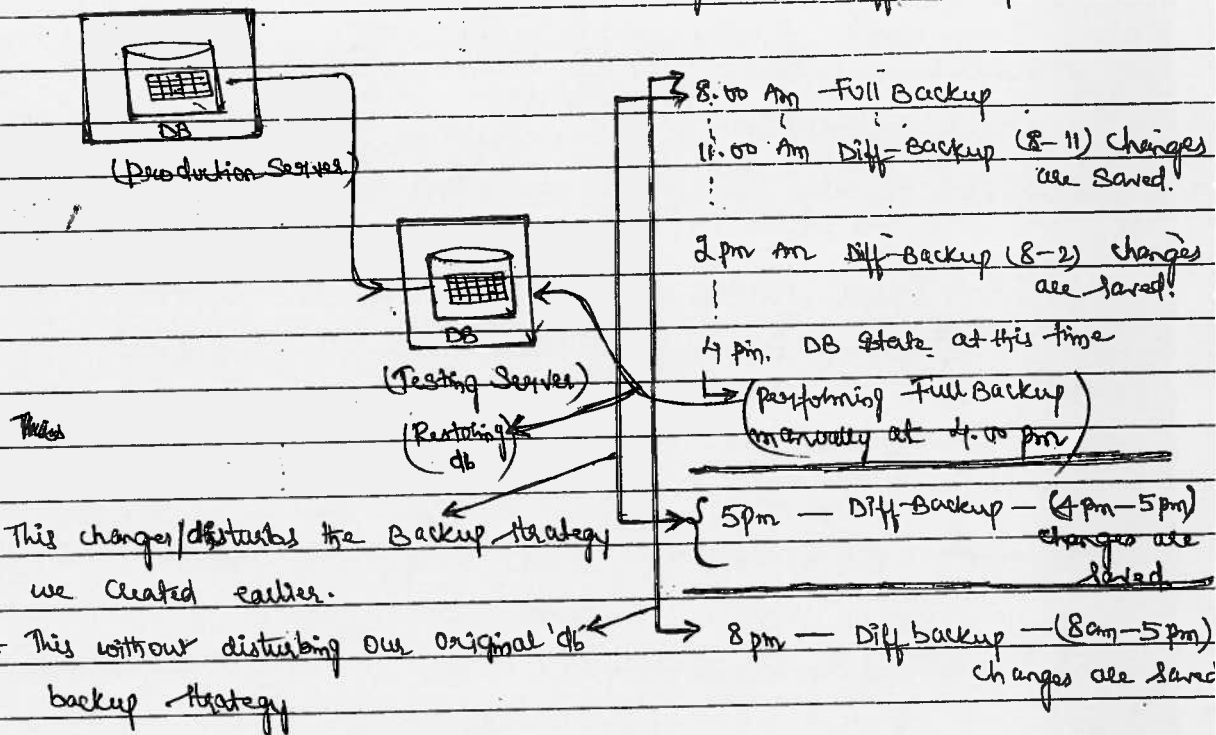
3. Copy only Backup :

- * This allows to take the 'db' backup (Generally Full Backup) without disturbing the existing backup strategy chain.
- * Generally, 'copy only backup' is used to copy the 'db' from one server to another server. (production server to Testing server).

Backup Strategy.

* Daily 8.00 Am Full Backup

* Every 3 hrs Diff-Backup



* This changes/disturbs the Backup Strategy we created earlier.

* This without disturbing our original 'db' backup strategy

-- T-SQL Code to perform Copy only Backup

```
Backup Database Testdb
To Disk = 'C:\Testfile.bak'
with Copy-only, password = '@peers123'
```

-- Restoring and decryption.

Restore Database Testdb

From Disk = 'C:\Testfile.bak'

with Replace, password = '@peers123'

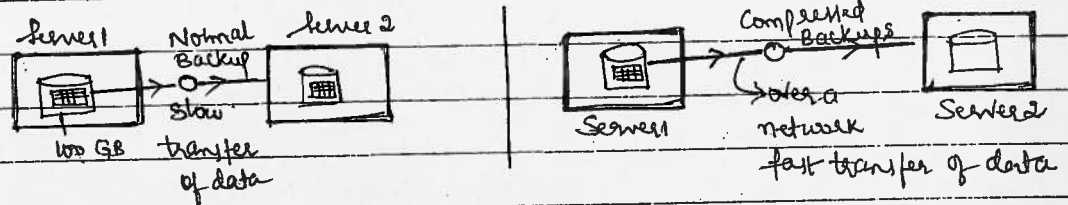
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6. Compressing Backups: SQL Server 2005 — 3rd party tools for Compression
 No Backups Compression tools
 SQL Server 2008 — DB Built-in Compression tool.

* This type of backups will save the storage space on backup media.

* In SQL Server 2005, 3rd party tools are used for Compression.

* In SQL Server 2008, Built-in db backup Compression exists.



* Generally 'Compressing Backups' will be used to transfer the database backup file across a network

Scenario 8:

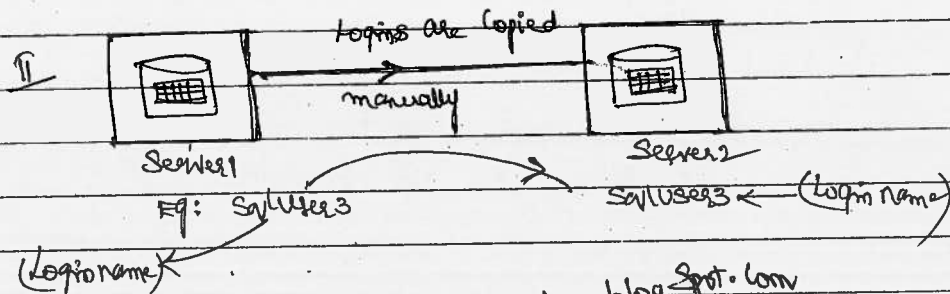
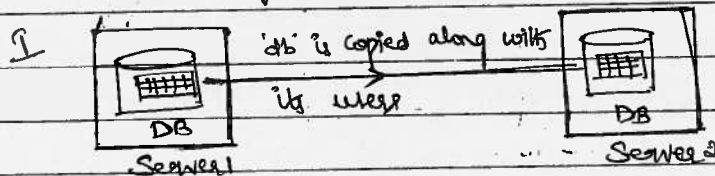
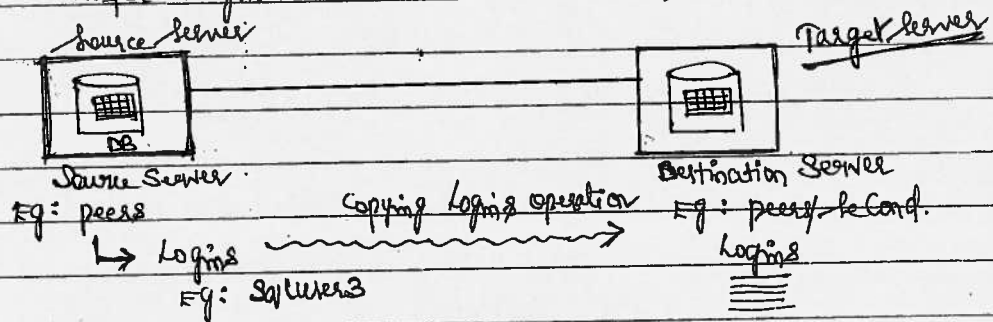
Q. How to copy logins from one server to another server?

(Windows / SQL Server Authentication login)

We have 2 methods to copy logins

① Through / using T-SQL Code method.

② Transfer Logins Task in SSIS (Method)



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- Sqlmail

- msdb

↳ hystables

DATE

SMTP Server } mail server (52)
Exchange Server

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DATABASE MAIL CONFIGURATION

* In SQL Server 2000, we have 'Sqlmail'

↳ COM based Api (Application programming Interface)

↳ MS-Outlook is must.

* In SQL Server 2005, we have

SMTP ← ↳ Sqlmail / Database mail (integrated mailing)

(implementation)
Transfer
protocol

↳ .Net Based Api.

↳ MS-Outlook is optional.

↳ part of IIS

Internet Information
System

SMTP Server:

* IIS is a webserver which is a part of windows operating system.

* IIS Contains a lot of services preferably as follows.

↳ web-site Management service

↳ FTP Service

↳ SMTP Service

↳ MSMQ Service - (Microsoft Message Queue Service)

To Install IIS:

Start Menu → Control panel → Add/Remove Programs

click on 'Add/Remove Windows Components'

check ☒ Application Server

click on ☐ Details button

check ☒ IIS click ☐ OK

check ☒ Ancillary services.

click ☐ Next After checking the required services.

Insert the windows (2003/2008) server O/S cd/dvd and

click ☐ Install

For Verification: * open IIS Manager

Start → Run → inetmgr

* To Give/view 'SMTP' details on our machine.

Start → programs → Admin Tools → pop3 service

* Mail Server Name = Peers

* Domain = dba.com

* Portno = 25

* Eg: Mail Accounts

hadi@dba.com

madhav@dba.com

classmate

PAGE

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** Configure System parameters

Account Retry Attempts : 3

Logging Level : Extended

Account Retry Delay : 120

click

click

Step ③ : Link SQL Server Agent with Mail profile (Mail server)

In Object Explorer

↳ Right click on 'SQL Server Agent'

↳ Select 'Properties'

Select 'Alert System' page

↳ Mail Session

/ check ☒ Enable Mail profile : Database Mail

Mail profile : SeniorAdmin profile

click

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Creating a Job : (Sunday : Full Backup)

In Object Explorer

↳ Select 'SQL Server Agent'

Right click and select 'Job'

Right click on 'Job' and select 'New Job'

** In 'New Job' window

Select 'General page'

Name : FullBackup Job

Owner : DBA Administrator

Category : Database Maintenance

Description : Full Backup of 2 or more databases

**

Select 'Steps Page'

click button

Step Name : Step1

Type : Transact-SQL script (T-SQL)

Database : Testdb

Command : Backup Database Testdb

To Disk = 'C:\Testdbfull.bak' ^{newshift}
with Name = 'Testdbfull', ^{existing} init
Go

**

Select 'Advanced page'

on Success Action : Goto the Next Step

on Failure Action : ✓ quit the Job Reporting failure
• Goto the Next Job

T-SQL :

Output File : Any file to record the step output

D:\Step1.txt

...

click

Again, click

Step Name = Step2

Type = T-SQL script

Database = Salesdb

Command = Backup database Salesdb

To disk = 'C:\Salesdbfull.bak'

with Name = 'Testdbfull', init

Go

* Verification of Jobs information.

use msdb

Go

Select * From SysJobs

Go

Select * From Sysoperators

* Executing a Job Manually.

In Object Explorer

↳ Select 'SQL Server Agent'

↳ Select 'Jobs'

↳ Select a particular Job (Eg: FullBackupJob)

/ Right click on the Job and

Select 'Start Job'

x (click 'Start') x

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3. MS db : (sysJobhistory)

-- T-SQL Code to know Job history / Status

Use MS db

Select * From sysJobs X

inner Join sysJobhistory Y

on X.Job-id = Y.Job-id

and X.name = 'Full Backup Job' ← Jobname

Note: To modify an Existing Job

In object Explorer

↳ Expand 'SQL Server Agent'

↳ Select 'Jobs'

↳ Select a particular Job (Eg: Full Backup Job)

Right click on it and Select 'properties'

Modify the Steps / Schedules using

New

Insert

Edit

Remove

In Backup Strategy Automation

Method 2. (Database) Maintenance plan wizard.

In object Explorer

↳ Expand 'Management'

↳ Select 'Maintenance plan'

Right click and Select 'Maintenance plan wizard'

Click Next

** Select a Target Server: (Eg: Peers)

Name: FullBackupTest

Server: Peers

Click Next

** Select Maintenance Tasks

☒ Full BackupClick Next

** Define full Backup Database Task

Databases: Specific Databases

Backup Type: full

Backup To: ☒ Disk☐ Tape☒ Create a Backup file for every database.

classmate {Folder: Home dir | mssql | Backup}

Note:

PAGE

DATE

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Verifying Notifications: (E-mail)Method I : * open any mail client Eg:outlook Express, MS-outlook,
web-based client etc,

* Connect with the Mail Account

* Check the inbox

Register the mail Account with MS-outlook as:

Open 'File' Menu → Select 'identity' → Select 'Add New Identity'

After Registration

Again open 'File' Menu → Select 'identity'

Provide 'Mail Account' details and

click OK

Method II : * Connect to SQL Server (as SQL Server Administrator)

use msdb ← (mailhost db)

(Mailing Related
Tables info)

Select * From Sysmail-log

Select * From Sysmail-mailItems

Select * From sysnotifications

* To verify the 'Job Steps' output files

full backup Jobs — p915 D:\Step1.txt

D:\Step2.txt

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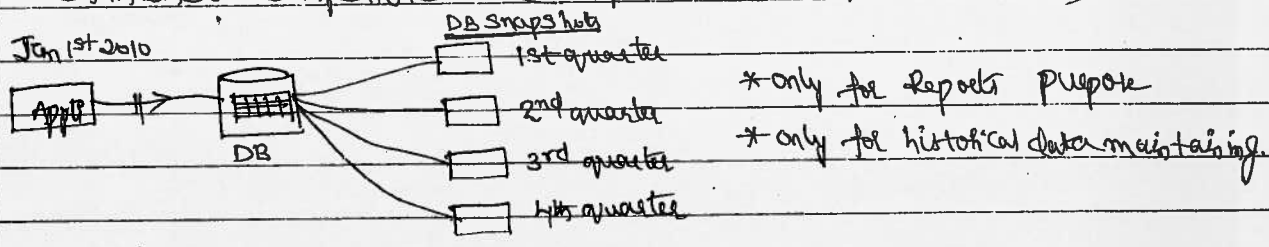
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To implement High Data Availability and failover, SQL Server provides 4 Methods.

1. Log Shipping } only Manual failover is possible
2. Replication }
3. Database Mirroring. } Automatic & Manual failover
4. Clustering services } is possible

DATABASE SNAPSHOTS : (New feature in SQL Server 2005)



* DB Snapshot means an image of a database created at a particular point of time.

* DB Snapshot are always Read-only & always static.

* DB Snapshot works based on 'Sparse files' technology.

* One or more db snapshots can be maintained for a database.

* Generally, DB Snapshot are also called as 'Snapshot Backup' of a 'db'.

Syntax: Creating Database Snapshot

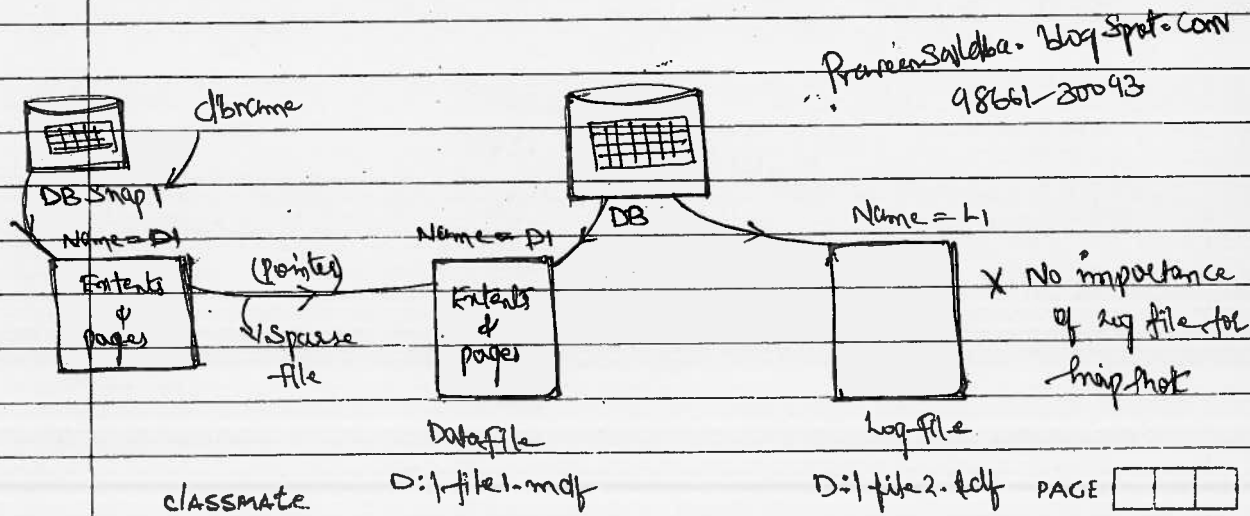
```

Create Database [Snapshot dbname]
ON ( Name = 'Logical Name'
    , FileName = 'Any path for mdf file'
    AS [Snapshot] of [Original dbname]

```

In DB Snapshot
log files have
no importance

Note: 'Logical Name' - it must be original db file logical name.



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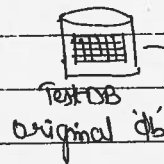
* Practical : How to Restore / Revert back the Original DB from db snapshot?

-- T-SQL Code to Restore a db from db snapshot.

Restore Database Testdb

From Database_SnapShot = 'Snap1at845am' ← db snapshot name

Jan 1st, 2016



- DB Snapshots
- ☐ -- 1st quarter -- dbsnap1 → delete
 - ☐ -- 2nd quarter -- dbsnap2 → delete
 - ☐ -- 3rd quarter -- dbsnap3 → delete
 - ☐ -- 4th quarter -- dbsnap4 → Latest db snapshot has to restored.

and all previous db snapshots should be deleted.

Note: we can Restore a db from latest db snapshots and --
all other (Existing) db snapshots should be deleted to perform --
restoring operation.

-- TO Drop a db snapshot

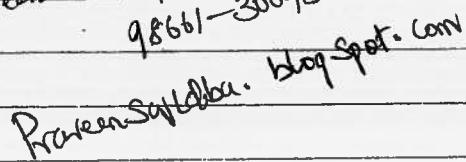
Eg: Drop Database Snap1at845am.

-- Press F5.

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Hardening process



* **Step 1:** Enable 'Database mirroring' feature on primary Server, witness Server and secondary server using Trace flags.

↳ open S.C.M

↳ Select a particular Sql Server instance

Right click on the 'MSSQL' Server instance.

Select 'properties'

Select 'Advanced' option

Select 'Startup parameters'

Startup parameters: ;-T1400

Note: T - Trace flag.

1400 - db mirroring feature trace flag no.

Note: Append ;-T1400

click Apply

Finally, click on OK button

Re-Start the 'MSSQL' Server instance.

Note: Similarly, append ;-T1400 in Startup parameters of secondary server and witness server and Re-start the respective Sql Server instances.

* **Step 2:** Maintain initial copy of Original 'db' on 'mirror Server' - as a mirror database.

For this, Use 'Backup & Restore' or 'Copy Database wizard' Method.

* Connect to 'Principal Server'

-- Verify the 'Recovery Model' of principal 'db' - Recovery model - Full

-- Take 'Full Backup' & 'Transaction Log Backup' of Original 'db'.

Eg: Testdb - Backup file path - D:\Testdb.bak

Here we take 'Full' & 'T-log Backup' into the same backup file.

Note: Right click on the Required db and Select the 'properties'

* Connect to Mirror Server.

↳ Restore the above backup file here

Note: while Restoring the 'db' - change the 'db' file path locations on the target server.

**** Include witness Server**

Do you want to Configure security to include a witness server instance?

☒ Yes

☐ No

click **Next**

**** choose Servers to Configure**

Save the security Configuration on the following :

☒ Principal server instance

☒ Mirror Server instance

☒ witness Server instance.

click **Next**

*** 'Principal Server Instance' page**

Principal Server instance :

Peers

▼

Listener port :

5022

☒ Encrypt data sent through this endpoint.

Endpoint Name :

princEndpoint

Note: If the principal, mirror and witness are instances on the

(a) same Computer/Server, their endpoints must use different ports.

(b) different Servers, the endpoints can use the same port-no. (5022).

click **Next**

**** 'Mirror Server Instance' Configuration**

Mirror Server Instance :

Peers/Second

▼

Connect

Listener port :

5023

Endpoint Name :

MirrorEndpoint

click **Next**

**** 'Witness Server Instance' Configuration**

Witness Server Instance :

peers/Third

▼

Connect

Listener port :

5024

Endpoint Name :

WitnessEndpoint

click **Next**

**** 'Service Accounts' - Configuration.**

Specify 'Service Accounts' for the following 'instances'

Principal :

dba/administrator

Witness :

dba/administrator

Mirror :

dba/administrator

click **Next**

classmate Complete the wizard window appears

Finally, click **Finish**

Note: In SQL Server 2008, No service packs required. (63)

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* Monitoring Database Mirroring:

We have 2 ways.

① DMV's — Dynamic Management Views

② Database Mirroring Monitor Tool

1. DMV:

→ Principal Server / Mirroring Server

sys.database-mirroring

sys.database-mirroring-endpoints

→ Witness Server

sys.database-mirroring-witness

sys.database-mirroring-endpoints

is-suspended — 0 — Synchronization
1 — not in synch'n.

↓ Column name
partners information

State-desc — Started synch'n
— Hopped not synch'n.

→ List of DB's configured for database mirroring.

-- T-SQL Code

Eg: Select db-name (database-id), *

from sys.database-mirroring

where mirroring-guid is not null.

2. Database Mirroring Monitor Tool (In SQL Server 2005, DB Mirroring Monitor Tool is not available)

In Object Explorer

→ Connect to principal server

→ Right click on 'Principal db'

→ Select 'Tasks'

→ Select 'Launch DB Mirroring Tool'

Click on 'Register Mirrored Database' option

Server Instance:

Click on button

Connect to the Required Principal / Mirror server

Select 'Testdb (synchronised, Peers/Second)'

Status:

⋮

⋮

⋮

⋮

⋮

Principal Log

Mirror Log

Unsent Log:

Unrestored Log.

⋮

⋮

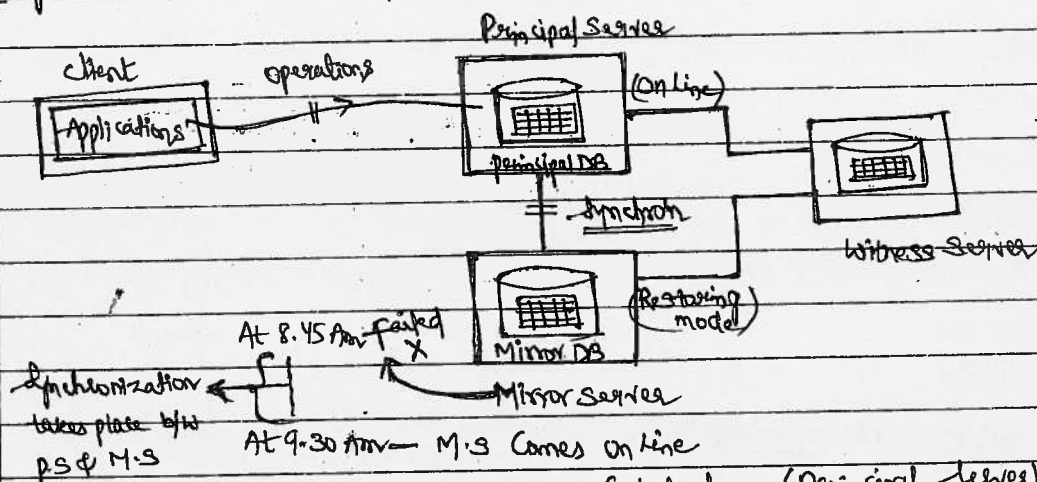
Operating Mode: High Safety with automatic failover (Synchronous)

classmate

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Testing : Automatic Fail Back

- ① Start the Principal Server.
- ② Stop the Mirror Server.
- ③ Re-start the principal server and Refresh.
- ④ Start the mirror Server.

Possibility 2. Mirror Server failed.

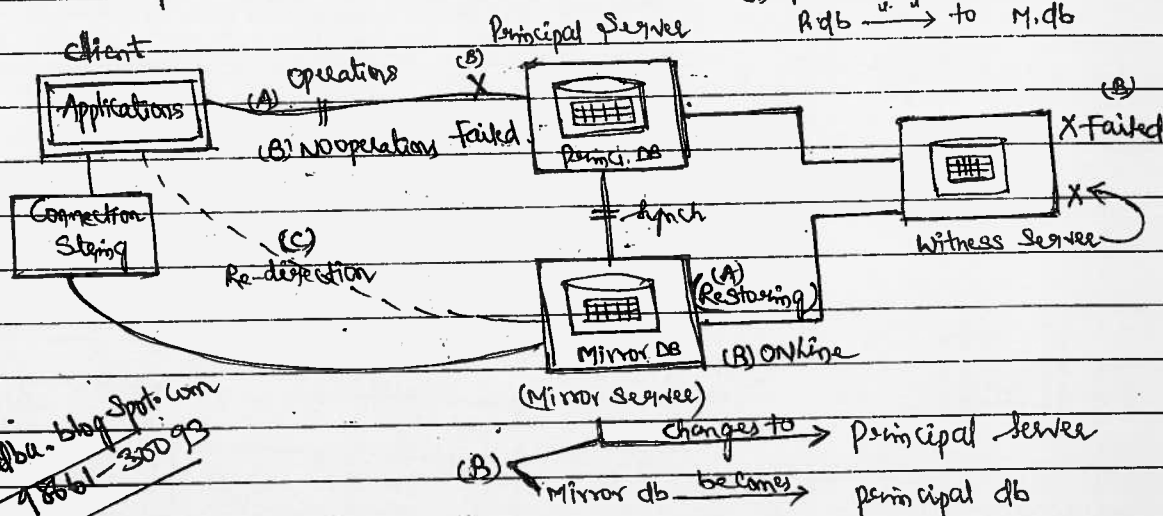
At 8:45 AM - failed
 At 9:30 AM - M.S Comes On Line
 Synchronization takes place b/w P.S & M.S
 means the transactions after 8:45 am are Copied from (Principal Server) to (Mirror Server)
 and the mirroring db will continue to stay in Restoring state.

- * Witness Server Confirms that Mirror Server is failed.
- * Applications users will continue the operations on principal db but synchronization between principal db & Mirror db will be stopped.
- * Once Mirror Server comes into online / Comes up, the synchronization will be automatically continued from the point of failure.

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Possibility 5. Principal Server & Witness Server failed. (c) p.s → M.S
Rdb → " " → M.db



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* In this, Application users cannot continue operations on principal server's principal db.

* We have to perform Manual failover. [and bring the Mirror Server's - mirror db into online. so that application users can access the db.]
By doing so, the applications are re-directed to Mirror server, which becomes principal server now, and the db present in it becomes 'principal db'.

Practical: How to bring the Mirror db online manually? when W.S X failed.

Step 1 Connect to Principal server and stop the 'DB' mirroring.

-- Testing purpose

Alter Database Testdb } this stops db mirroring of principal db
Set partner off

Step 2: Case (i) if possible Connect to 'Principal Server'

Take Tail-Log Transaction Backup

-- On Principal Server

Backup Log Testdb

TO Disk = E:\Testdbtail.bak'

with No-Truncat

Eg: Testdb

Step 3: Restore this Tail Log Backup on Mirror Server - mirror db
with 'Recovery' option

This brings the DB (Testdb) into 'Online' state

-- ON Mirroring server

Restore Log Testdb

From Disk = 'c:\Testdbtail.bak'

with Recovery (db comes into online)

DATE

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-- Steps to create a database snapshot on a microsoft database.

Step ① : Stop the principal server

Step ② : This will bring the 'Mirror db' into 'on line' and will be promoted as principal db. (on Mirror server)

Step ③ : Create the DB Snapshot (for the promoted 'principal db')
Verify the 'DB' - datafiles logical names.

Note : Snapshot should be created on the same instance.

Verify db - datafiles logical names. using the following hint.

* Exec sp_helpdb Testdb

create database snap

on (Name = 'Testdb',

File Name = 'c:\snap1.mdf'

AS Snapshot of Testdb

Step ④ : Perform Automatic failback process.

Note : DB Snapshot are created on mirror server location to decrease the load on physical server location.

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LOG SHIPPING

* The process of shipping the transaction log backups from one server db to another server db is called 'log shipping'.

* log shipping is implemented per-database basis.

* log shipping Requires.

① Primary server

② Secondary server

③ Monitor server (optional)

Recovery Model of

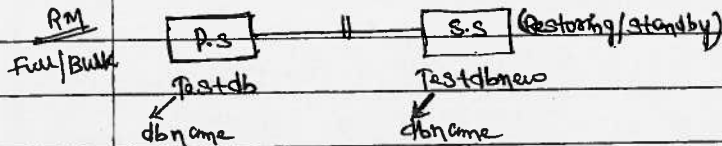
• Primary Server

→ Full/Bulk logged

• Secondary Server

→ Restoring/Standby

• Any no. of Secondary servers can be used.



only Monitoring Purpose
generates Reports
No Automatic failover

Note: 'DB' names need not be same in log shipping.

1. Primary Server: A server which maintains original db is called primary server.

* This database is called 'primary database'.

* Recovery model of primary db may be full/Bulk logged.

2. Secondary Server: A server which maintains copy database of 'principal db' is called secondary server.

* This 'db' is called 'secondary db'.

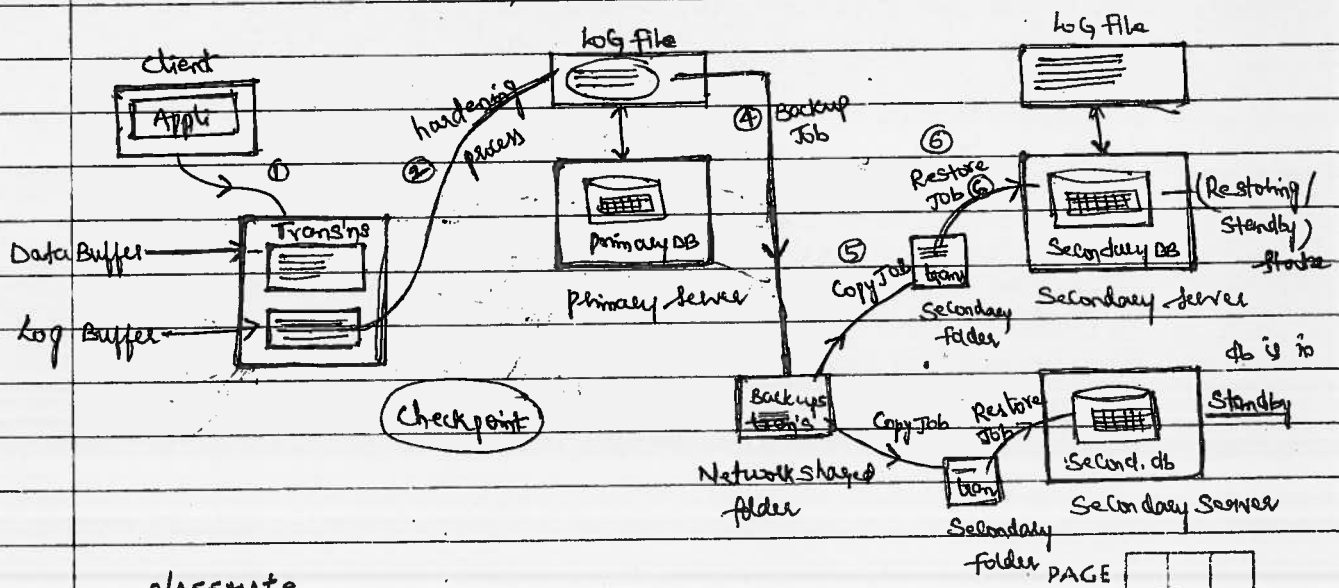
* Recovery state of secondary db may be: ✓ standby

* One or more secondary servers can be configured with secondary server - secondary databases.

3. Monitor Server: A monitor server provides

(a) Monitoring of primary & secondary servers

(b) Alerts Generation.



DATE

Step 3: Configure the Primary Server 'DB' for Log Shipping.

Connect to 'Primary Server' (Eg: Peers)

Select a primary db (Eg: DWHDB)

Right click on the 'db' and

Select 'Properties'

'DB properties - DBName' window appears

Select 'Transaction Log Shipping' option.

check

☒ Enable this as a primary db in Log Shipping.

T-Log Backups:

Click on Backup Settings

** Transaction Log Backup Settings - window appears

Network path to the Backup folder

/ \\Peers\primary folder

Provide local folder path

C:\primary folder

Backup Job:

Job Name: LS-Backup-DWHDB

Schedule

* Schedule Every 30 minutes

* Delete files older than: 48 Hours ▼

* Alert if no backup occurs within: 40 minutes

Finally, click OK

** Secondary Servers & Databases:

click Add button

'Secondary DB Settings' window appears

Secondary Server Instance:

Connect

click on Connect to required Secondary Server instance.

In Initialize Secondary database options:

☐

☐

✓ ☒ NO, the Secondary db is initialized.

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Testing Log Shipping process:

- ① Connect to primary server db
- ② perform some operations on primary db of primary server
- ③ verify the operations results in secondary db after the scheduled DB time.

* Manual failover in Log Shipping.

- Step ① Manually Corrupt the primary db [datafile] If: Rename the db file name delete mdf file
- To do this, Stop the Primary Server using SCM and delete the datafile of the primary db (delete .mdf file of the db).
- Step ② On Secondary Servers:
- ②a Run the 'Copy Job' Manually on Secondary server
This Copies the 'tar' files from primary server folder (network shared folder) to Secondary folder which are not copied earlier.
- ②b Run the 'Restore Job' Manually on Secondary Server
This Copies/restores the 'tar' files from 'Secondary folder' into the Secondary database which are not restored earlier.
- Step ③ Perform 'Tail log' Transaction Backup on primary Server (this backs up the active transactions' which are not backed up earlier).
change the 'Primary db' state to 'Restoring / Standby' state
-- Perform Tail log Backup
- { Backup Log DWTDB
 To disk = 'c:\dwhtdbtaillog.bak'
 with No-Tuncate, NoRecovery ← db comes into Restoring mode.
- Step ④ Now Perform Restoring of Tail log on Secondary server with 'Recovery' mode ← This will bring the secondary db online.
-- Restore operation
- { Restore Log DWTDB
 From Disk = 'c:\dwhtdbtaillog.bak'
 with Recovery ← db comes into 'Online'
- Note: Inform this information to Application Manager so that he will modify/change the connection string information.

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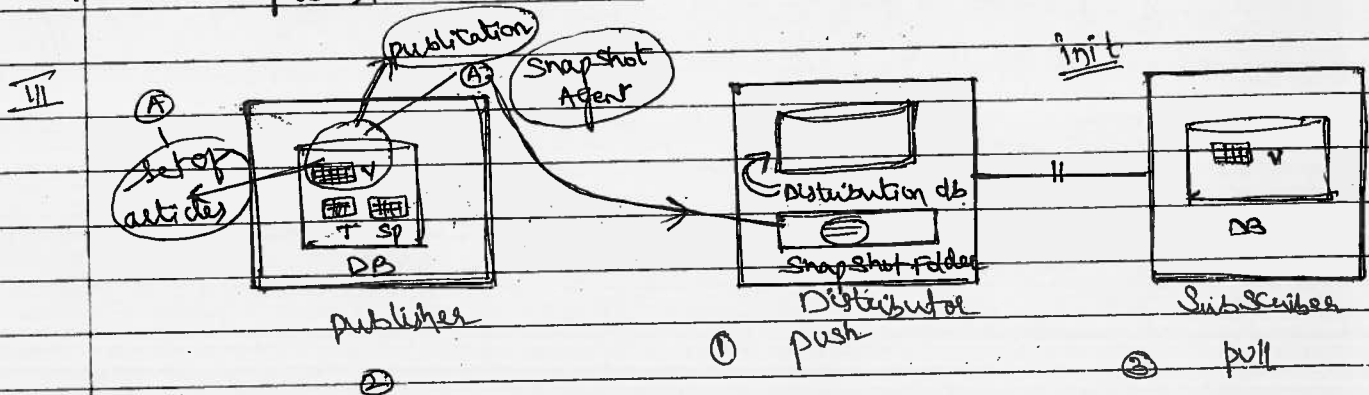
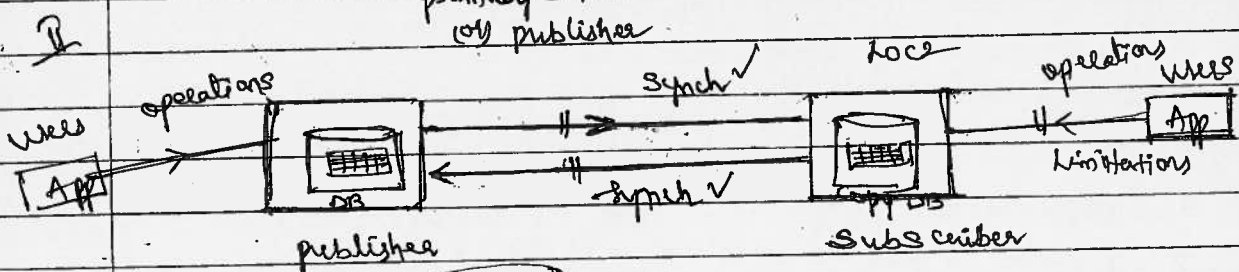
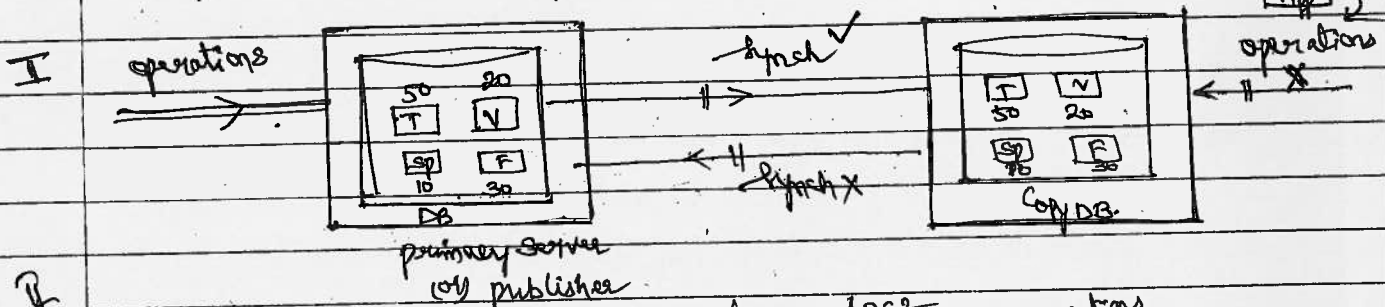
REPLICATION

- * Sql Server db Replication is a set of technologies for copying and distributing data and database objects from one db to one or more other db's and keeping the db synchronized.
- * Using Replication, we can increase the availability of our IT solutions by geographically distributing our data, making applications more - resilient to communication failures.
- * Replication is used to distribute the data-access workload across multiple servers or to support remote clients.
- * Replication is an automatic migration of data and db objects from one server db to another server db.
- * Replication is implemented per 'db' basis or a set of db objects in the database.

↳ Database level

↳ Objects level

- * Replication provides 'High Data Availability' and 'Manual failover' uses



[P] — publisher
 [D] — Distributor
 [S] — Subscriber

DATE

1. Distributor : This is an intermediate server which maintains

- * distribution db
- * Snapshot folder
- * Publisher & Subscriber(s) information.
- * replication metadata

2. * Responsible for managing the distribution db, which holds replication status data, metadata, and the actual data that sql server replicates.

Note : ① A single 'db' server instance can act as both Distributor and publisher, here Distributor is called a 'local Distributor'.

② when we configure the [P] and the [D] on two separate 'db' servers instances, the [D] is called Remote Distributor.

2. Publisher :- A server which maintains original db.

- original owner of the inform'n that is published.
- one or more publications can be created for a 'db'.
- 'Publication' means set of articles (Eg: tables, views, sps etc)

3. Subscriber : A server that receives copies of the publication and provides the data to end users & applications.

Note: In some configurations

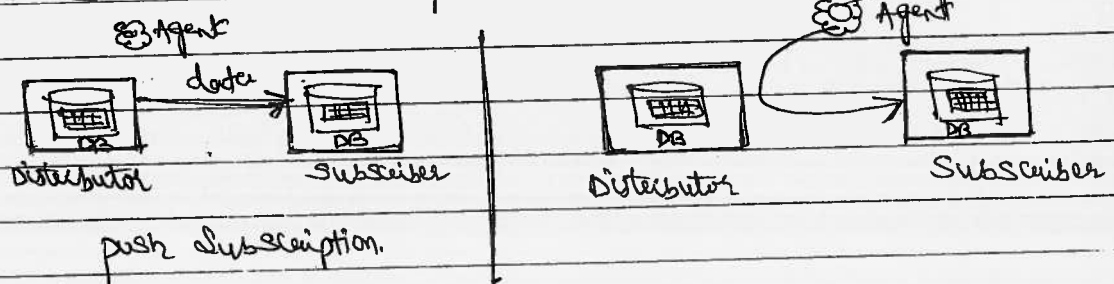
- ✓ Subscribers have a read-only copy of the 'db' and in other
- ✓ we can update information in the subscribers and replicate those changes back to the publisher.

Subscriptions are of 2 types

① push Subscriptions : Distributor / (Distributor Agent) Copies the data to the Subscriber db

② pull Subscription : Subscriber retrieves the data from the Distributor

(Note: Distributor Agent Runs on Subscriber)



DATE

Eg (Snapshot)

Eg (Snapshot)

SETTING UP A REPLICATION OR STEPS TO IMPLEMENT A REPLICATION.

1. Setup a Distributor for the publisher to use. (eg) Configure the ☐ ☐ ☐ ☐
2. create a publication on Publisher to replicate that includes the articles we want to copy.
3. Configure the Subscriber(s) and Subscription(s).

- (eg)
- ① Configure the Distributor
 - ② prepare the publisher and publication
 - ③ prepare the Subscriber(s) and Subscription(s).

I. Configuring Snapshot Replication / Implementing Snapshot Replication.

publisher — Peer

/ Distributor — Peer/Third

Subscriber — Peer/Second.

* Step 1: Setup the Distributor / Configure the Distributor.
(Configuring the Distributor as a Remote Distributor)

↳ open SSMS

↳ Connect to the db engine instance that we want to configure as distributor
(Eg: Connect to Peer/Third)

↳ Right click on 'Replication' folder and
choose 'Configure Distribution'

↳ On 'Configure Distribution wizard' page
click ☐ ☐

* * 'Distributor' page

✓ ☐ Peer/Third will act as its own Distributor.

* ☐ Use the following Server as the Distributor.

☐ Add

click ☐ ☐

* * 'Sql Server Agent Start' page

✓ ☐ Yes Configure the Sql Server Agent service to start automatically.

click ☐ ☐

* * 'Snap Shot Folder' page

Snapshot folder : c:\program Files\Microsoft SQL Server\MSSQL.1\MSSQL\Repldata

click ☐ ☐

*

Step 2: Prepare the publisher and publication. (01)

creating a publication on a publisher to replicate that includes all the articles we want to copy.

↳ Open SSMS.

↳ Connect to the db engine instance in which we want to create 'Publication'
(Eg: Peers)

↳ Expand the 'Replication' folder

↳ Right click on 'Local publications' and
choose 'New publication'

↳ on 'New publication wizard' page
click **Next**

* 'Distributor' page

○ 'Peers' will act as its own Distributor

✓ ○ Use the following server as the Distributor.

Peers/third

Add

click **Add**

Connect to Server : **Peers/third** ← Distributor Server

click **Connect**

click **Next**

* 'Administrative password' page

Specify the pwd. required for 'Peers' to establish an admin. link with
the Distributor, Peers/third.

password : **@peers123**

Confirm pwd : **@peers123**

click **Next**

* 'publication Database' page

Databases :

- Adventure

- Testdb ✓

Select 'Testdb' for Eg:

click **Next**

* 'Publication Type' page

publication Type : ✓ Snapshot publication (Select this option)

Transactional "

Transactional " with updatable subscription

Merge Publication

classmate

click **Next**

DATE

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*** 'wizard Actions' page

☒ Create the publication

☒ Generate a Script file with steps to create the publication.

click **Next**

→ file name: **c:\publisherScript.sql**

*** 'Complete the wizard' page.

Click Next

publication Name: **Snapshot demo**

* 'Creating publication' page

Shows the result of the process done.

click **Close**

Verification: ① Goto Snapshot folder path on (Peers/third) server

<HD>\MSSQL\repdata\unc\Peers-Testdb-Snapshot demo\20110129101948

yr mth date

• bcp — bulk-copy data file

• idx — index script of article

• Pre — Snapshot Script

• Sch — article schema

for Each article 4 files are Created.

② Goto publisher Server eg: peers

Expand 'Replication' folder

Expand 'Local publications'

Here we see '[Testdb]: Snapshot demo' is Created.

*

Step 3: How to Subscribe to the publication (or)

Prepare the Subscribers and Subscriptions

→ Open SSMS.

→ Connect to the 'Subscriber' db Engine instance (eg: peers/second)

→ Expand 'Replication' folder

→ ~~Right~~ Right click on 'Local Subscriptions' and choose 'New Subscriptions'

on 'New Subscription wizard' page

click **Next**

*** 'publication' page

Publisher:

peers ☒

<Find SqlServer publisher>

<Find Oracle Server publisher>

Select/Choose 'Find SqlServer publisher'

classmate Connect to Server: **Peers**

click **Connect**

PAGE

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'Distribution Agent Security' window page

○ Run under the following windows account.

✓ ○ Run under the Sql Server Agent Service Account. (Select this option)

Connect to the Distributor

○ By impersonating the process account.

○ Using a Sql Server login.

Connect to the Subscriber

○ By impersonating the process account.

○ Using the following SQL Server login.

click OK

click **Next**

** 'Synchronization Schedule' page

Agent Schedule:

Subscriber:

peers | teCond

Agent Location

Distributor

Agent Schedule:

▾

Run Continuously

Run on demand only

< Define Schedule >

Select 'Define Schedule'

Daily frequency: 1 hours to.

click OK

click **Next**

* 'Initialize Subscriptions' page

Subscription properties:

Subscriber

peers | teCond

initialize

☒

Initialize when

immediately

▾

At First synchronization

Select 'immediately' option

click **Next**

** Wizard options.

☒ Create the Subscription(s)

☒ Generate a Script file

click **Next**

File Name: C:\SubscriberScript.sql

click **Next**

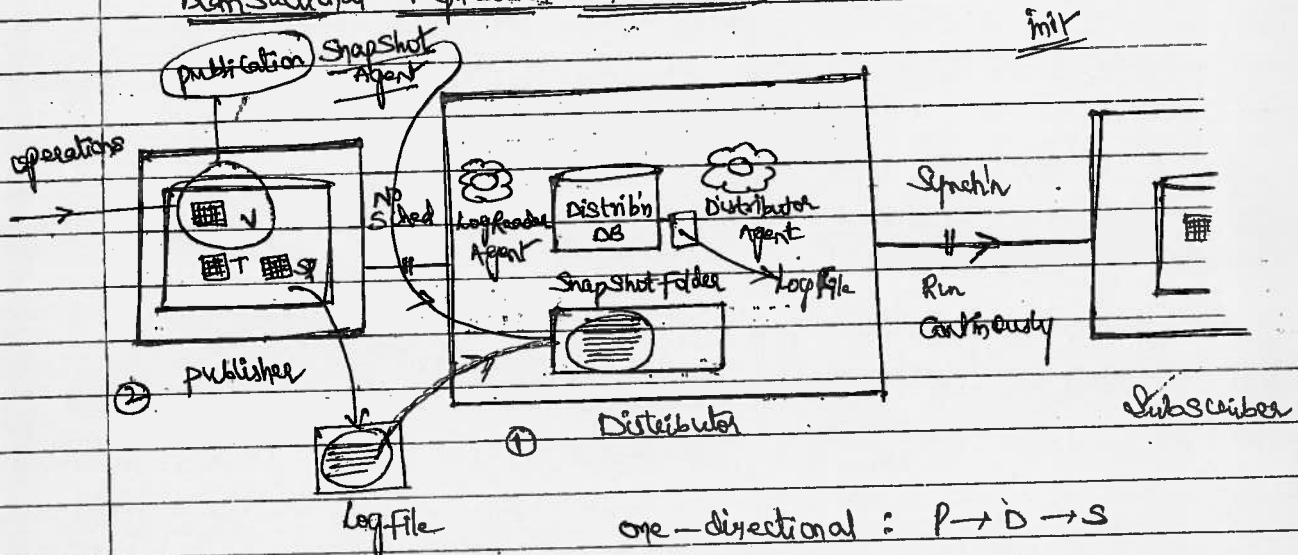
* Complete the wizard.

click **Finish**

Transactional Replication Implementation process

- ① & ② are same as of Snapshot process.
 - ③ Transactional Replication is one-directional and immediately Publishers changes are immediately synchronized to Subscribers.
 - ④ Operations performed in the publisher's db are taken by the Log Agent continuously and written into Distribution Log.
 - ⑤ Distribution Agent will copy these transactions on subscribers as per synchronization schedule. (Generally, Run Continuously)
- Note: A table must have a primary key while performing TRS

Transactional Replication Architecture



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* * 'Snap Shot Agent' page

- ☒ create a Snap Shot immediately and keep
- ☒ Schedule the Snap shot Agent to run

click

* * 'Agent Security' page

Snap Shot Agent:

☒

{ click on 'Security Settings'

In 'Snap Shot Agent Security' window

☐ Run under the ~~all~~ windows account:

process account:

password:

Confirm pind:

☒ Run under the SQL Server Agent service Account.

Connecting to the publisher:

☐ By impersonating the process account.

☐ Use the following SQL Server login

login:

pwd:

Confirm pind:

click

click

* * 'Wizard Actions' page

☒ create the publication

☒ Generate a Script...

* * 'Script file properties' page

File Name:

click

* * 'Complete the wizard' page

publication Name: Transpubl

click Finish.

* * 'creating publication' page

click

Verification: Goto 'Pers' Server

↳ Expand 'Replication'

↳ In 'Local publications'

'Transpubl' publication —
is Created.

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click ☒ in Subscription db.

< New db. >

< Testdb >

choose 'Testdb' db

Note: initially, we can add any no. of subscribers.
click

* 'Distribution Agent Security'

click on in 'Connection to Subscriber' column.

on 'Distribution Agent Security' window.

○ Run under the full windows account.

○ Run under the SQL Server Agent Service Account.

— Connect to the Distributor

— Connect to the Subscriber

click

Subscription properties:

Agent for Subscriber

peers | Secnd.

Conn. to Distributor

Impersonate process Acc

Conn. to Subscriber

Imp. process Acc

click

* 'Synchronization Schedule' page.

Agent Schedule:

Subscriber

peers | Secnd

Agent

Distributor

Agent Schedule

☒ Run Continuously

Run on demand only

< Define Schedule >

click

* 'Initialize Subscriptions' page

Subscription properties:

Subscriber

peers | Secnd.

Initialize

☒

Initialize when

☒ Immediately

At first synchron

click

* Wizard Actions

☒ Create the Subscription(s)

☒ Generate a Script file.

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Script file properties

File Name =

click

C:\Transdb1.sql

PAGE

- peers — publisher
- peers/third — distributor
- peers/second — subscribers.

DATE

(79)

II Configuring the Merge Replication / Implementing Merge Replication process.

* Step 1: Setup the Distributor / Configure the distributor
(Eg: peers/third)

Follow the Step 1 of Snapshot Replication to Config distributor.

* Step 2: Create a publication (Eg: Mergepub1) on a publisher (Eg: peers)
to replicate data which includes all the articles. that we want to copy.

(or) Prepare the publications and 'publishers'

↳ Open SSMS

↳ Connect to the 'db' instance server in which we want to

Create a 'Publication' Eg: Peers — Publisher.

↳ Expand 'Replication' folder

↳ R/c on 'Local publications' and

choose 'New publications'
on 'New Publication wizard'

click **Next**

* Publication Database page

Databases: Adventureworksdb

☒ Testdb

click **Next**

* 'Publication Type' page

- Snapshot publication

- Transactional "

- Transactional " with updatable subscriptions

☒ Merge Publications

click **Next**

* Subscriber Types

☒ SQL Server 2005

☐ " " " Mobile Edition.

☐ " " 2000

☐ " " for windows CE

click **Next**

* 'Articles' page

Objects to publish

☒ Tables — Eg: Amp2
dept2

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click **Next**

PAGE

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* Step 3: How to Subscribe to the publication?

by Prepare the 'Subscriptions' and 'Subscribers'

↳ open SSMS

↳ Connect to the 'Subscriber' server (eg: peers2cond)

↳ Expand 'Replication' folder

↳ R/C on 'Local Subscriptions'

choose 'New Subscriptions'

On 'New Subscription wizard' window

click Next.

** 'Publication' page

Publisher:

peers2

↳ <Find SQL Server publishers>

<Find Oracle Server publishers>

Select 'Find SQL Server publishers'

Connect to Server:

peers2

← Publisher

click Connect

to Publisher: peers2

Databases & publications:

☐ Testdb

Snaprep1

Transpub1

Mergepub1 ✓ (select this)

** 'Merge Agent Location' page

✓ push Subscription — (Agents run at Distributor)

○ pull Subscription — (Agents run at its Subscriber)

click Next

** 'Subscribers' page

click on Add SQL Server Subscriber

Connect to Server:

peers2cond

← Subscriber

click Connect

Subscribers & Subscription dbs:

Subscriber

☒ peers2cond

Subscription DB

☒ Testdb

Branch1

↳ <New DB>

<Testdb Branch1>

choose 'Testdb'

PAGE

Branch1

classmate If, we can add any. no. of Subscribers.

click Next

DATE

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*
** Complete the Wizard.click Finish*
** Creating Subscriptions.click CloseHandling Merge Conflicts.

1. Merge Conflict is a situation where 'publisher' users & 'subscriber' users updating / inserting the same records with different values.
2. Conflict Resolver will resolve the conflicts based on Subscriber type and Conflict Resolution Process (CRP)

Note: Subscriber Type

Client	→ Always	P	wins
Server	→ CRP	— ?	

Q. How to Handle Merge Conflicts Manually?

open SSMS

Connect to the Publisher Server (eg: Peer1)

Expand 'Replication'

Expand 'Local Publications'

Select 'Merge Publication' eg: Mergepubdemo

R/C on 'Mergepubdemo' and

choose 'View Conflicts'

In 'Conflict Resolver' window

Select 'publisher' Rows / 'Subscriber' Rows

click 'winner / Loser'

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Q How to Configure Replication Related Alerts to get Notifications?

↳ open SSMS

↳ Connect to 'Publisher'

↳ Expand 'Replication'

↳ Expand 'Local publications'

R/C on a particular 'Publication' (Eg: Transpub1)

Choose 'Launch Replication Monitor'

☐ Replication Monitor

☐ My publishers

☐ Peers

☒ Testdb: Snappub1

☒ Testdb: Transpub1

☒ Testdb: Mugepub1

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Select 'Transpub1' publication for ex.

Now Select 'Warnings and Alerts' tab in 'Replication Monitor' window

Warnings:

Enabled	Warning	Threshold
☐	☒	☒

click **Configure Alerts**

In 'Configuration Replication Alerts' window.

Replication Alerts: ☒

Choose the required Alert Message Eg: Replication: Agent Failure

click **Configure**

Now 'Replication: Agent Failure' alert properties — window comes.

In 'General' page

Name = Replication: Agent Failure ☐ Enable.

Type = ▼

Event Alert Definition:

Database Name: ▼

Alerts will be raised on:

☒ Error Number: 14151

☐ Severity

☐ Raise alert when Message Contains:

Message Alert:

click **OK**

DATE

--	--	--	--	--	--

Configuration of Transactional Replication with updatable Subscribers.

* Step 1: Setup the Distributor / Configure the Distributor.

Follow the step (I) of Snapshot Replication to Configure 'Distributor'.

* Step 2: Prepare the publisher and publication(s).

↳ open SSMS.

↳ Connect to the db engine server in which we want to create 'Publication'.

Eg: Peers.

↳ Expand 'Replication' folder

↳ R/c on 'Local Publications' and

choose 'New Publication'.

On 'New Publication wizard' Page

click Next

** 'Publication Database' page

Databases: ☒ Testdb☐ Loddudb☐ Advendb.

Select 'Testdb' database

click Next

* 'Publication Type'

Select 'Transactional Publication with updatable subscriptions'.

click Next

* 'Articles' page

Objects to publish:

Select the required articles Eg: Ampphone

click Next → (A), (B)

(B) { ** 'Filter Table Rows' page } X

Filter Tables:

--

Filter:

click Next

(A) ** 'Articles Issues' page

Issues: Unique identifier columns will be added to tables

click Next

* Step 3: How to subscribe to the publication? (4)

Prepare the Subscriptions and Subscriber(s).

↳ open SSMS

↳ Connect to 'Subscriber' server Eg: peers/1econd.

* ✓ (a) Subscriber must have a linked server to the Publisher (Peers)

Connect to the 'Subscriber' server Eg: peers/1econd.

↳ Expand 'Server Objects'

↳ R/c on 'Linked Servers' and

choose 'New Linked Server'

→ In 'New Linked Server' window

Select 'General' page

Linked Server: Peers

Server Type: ☒ SQL Server

☐ Other data source

→ In 'Security' page

Local Server login to remote login mappings:

Local Login	Impersonate	Remote user	Remote password

Add

Remove

For a login not defined in the list above, connections will:

✓ ☒ Be made using the login's current security context

Click

* ✓ (b) Prepare the Subscription database.

↳ open SSMS

↳ Connect to the 'Subscriber' server (Eg: peers/1econd)

↳ Expand 'Replication'

↳ R/c on 'Local Subscriptions' and

choose 'New Subscriptions'

on 'New Subscription wizard' page

click

* ** 'Publication' page

publisher: ▾

- Databases & publications:

☐ Testdb

☒ Transupdate Publi ✓

classmate

click

* * 'updatable Subscriptions' page

Replicate Subscriber changes:

Subscriber
peers | de Cond

Replicate

☒

Commit at Publisher

☒ Simultaneously Commit changes. ☒
Queue changes & Commit when possible.Click **Next**

* * 'login for updatable Subscriptions' page.

When Connecting to the publisher

○ Create a linked Server that connects using Sql Server Authentication.

Login: password: Confirm password:

defined.

✓ ○ Use a linked Server or remote Server that you have already—

/ Click **Next**

* * 'Initialize Subscriptions' page

Subscription properties:

Subscriber

peers | de Cond

Initialize

☒

Initialize when

Immediately ☒

At first Synch

Click **Next**

* * 'wizard Actions' page

☒ Create Subscriptions☐ Generate a Script fileClick **Next**

* * 'Complete the wizard' page

Click **Finish**

* * 'Creating Subscriptions' page

failure

Status is shown here Eg: Success/ Completed with errors/

Click **Close**

* ✓ (c) Enable Subscription 'ds' for updates.

↳ Sp-link-publication → Execute this on Subscriber(s)

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{ Peers — publisher
 Peers/third — Distributor
 Peers/second — Subscriber

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Configuration of peer-to-peer Replication

* Step 1: Setup the Distributor / Configure the Distributor Eg: Peers/third

* Step 2: Provides all participants information to the Distributor.

↳ Open SSMS

↳ Connect to the 'Distributor' Server

↳ R/C on 'Replication' and

choose 'Distributor Properties'

In 'Distribution Properties' — Peers/third window

Select 'Publishers' page

click

↳ Add Sql Server publisher

↳ Add Oracle publisher

choose 'Add Sql Server publisher'

Connect to Server:

click

Again click

Select or choose 'Add Sql Server publisher'

Connect to Server:

click

Publishers:

Publishers:

Distribution Database

☒ Peers

~~Testdb~~ Distribution

☒ Peers/second

~~Test~~ Distribution

click

Administrative link password:

password

Confirm password

click

Step 3: Maintain initial copy of database and required — articles on every participant

(Using Backup & Restore through T-Sql Script)

Eg: Peers — (publisher)

Database Name = Testdb

Take full Backup & Transaction Log Backup and Restore it on all participants with recovery state as 'with Recovery' ← Online

↳ peers/second.

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Peers and Peer databases :

Peer Server Instance

Peer Database

☒ Peers | Server2 Second

Branch2

☒ Peers | Server3 fourth

Branch3

Add Sql Server

click Next

*** 'Log Reader Agent Security' page

'Agent Security properties :

Agents for publisher peers Database

Connection to
Distributor

Connection to
Publisher

Peers | Second

Branch2

Imp. S.S.A. S. Acc

Imp. p. Acc

...

Peers | Fourth

Branch4

" "

" "

...

Peers

Testdb

" "

" "

...

click Next

*** 'Distribution Agent Security' page

Agent Security properties :

Agent for Subscriber Peer DB

Connection to
Distributor

Connection to
Subscriber

Peers | Second

Branch2

Imp. pub. acc

Imp. pub acc

...

Peers | fourth

Branch4

" " "

" " "

...

Note : Run under the SQL Server Agent Service Account

click Next

*** 'New peer Initialization' page

✓ I created the peer db manually.

○

Specify the backup file used

Browse

click Next

*** 'Complete the wizard' page.

click Finish

*** 'Building the Peer-to-Peer Topology'

click Close

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MONITORING PERFORMANCE AND OTHER ACTIVITIES

(Monitoring Database System)

- * Performance problems
- * Sql Server Level Configuration parameters.
- * Database Tuning
- * Query / Stored procedure Tuning
- * Operating System parameters.

Generally an Administrator will monitor

- * Applications (eg: Logins)
- * Network Resources
- * Sql Server
- * Databases
- * Objects

* Hardware Resources (Cpu, disk i/o, RAM utilization)

Monitoring is Required for

- ① Troubleshooting at Sql Server level & at db level errors.
- ② Tracking the users activity / users load.
- ③ Resolving performance problems and optimizing the performance.
- ④ Identifying the Connectivity issues / Bottlenecks.

Monitoring Tools are:

- ① Windows Event Log — O/S Level.
- ② Sql Server Event Log.
- ③ System Stored procedures
- ④ Task Manager — O/S Level.
- ⑤ Activity Monitor —
- ⑥ Performance / System Monitor — O/S Level.
- ⑦ Sql Server profiler
- ⑧ DMV's and DMF's (new in Sql Server 2005)
- ⑨ Dash Board Reports in SSMS

Third party Monitoring Tools are:

- ① Spot Light
- ② Fog Light
- ③ Sql TA (Trace Analyzer)
- ④ Sql DM (Diagnostic Manager)
- ⑤ SQL Life

www.quest.com (62)

classmate

www.idera.com (34, 5)

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* ~~SQL Server~~ Service Importance :

- * This service is a Common service for all instances. (instance Unaware Service).
- * This provides all instances information, Connectivity information to the clients.

2. Tracking the Users Activity / users load :

- * Every Application will use one or more logins to perform operations with db's in the db server.
- * All Operational users of that application will access the db's in the context of one or more logins. (Session)
- * For every operational users one (or more) process is created in the db server whenever a user connects.
- * The Session will be destroyed when the user Disconnects.
- * By default, Session memory is 1MB.
- * All the sessions are created in 'tempdb' database.

• To track the Users load / users Activity :

System Tables

↳ In Master db — Sysprocess ← table

* System stored procedures

(sp-who, sp-who2)

* DMV — Dynamic Management Views

↳ Sys.dm-exec-sessions

* Activity Monitor — GUID — (1-50 → spid — System processes
750 → Application users related processes)

SPID — Server process id

- * Every process will have Unique id.
- * System processes are from 1-50.
- * Users processes are from 51 to 750

(Applications) Connected to SQL Server means one or more logins are connected to SQL server.

Eg: sqluser1

sqluser2

Reference: www.sqlalchemy.com
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* db - database

* Sp - stored procedure

* SSMS - SQL Server Management Studio

* R/c - Right click

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* DMV

Select * From Sys.dm-exec-Sessions

This stored procedure returns same information as 'Sysprocess' SP + reads, writes, logical reads, etc.

* { GUID (mode) } Activity Monitor

open SSMS

In Object Explorer

↳ Connect to a particular server (eg: Pears)

↳ Expand 'Management'

↳ R/c on 'Activity Monitor'

↳ Select 'View process'

Note: Only user processes will be displayed but not system-processes.

* To view the SQL statements / SP's submitted by a particular process, we use: - dbcc inputbuffer(sp_id)

* To view the locks applied by the particular processes, we use Exec sp_lock sp_id.

Note: if sp_id is not specified then all processes locks will be displayed

3. Resolving Performance problems & optimizing the performance.

Performance of the db system is estimated based on two things:

(a) Response Time: Time taken for each transaction.

<u>Performance</u>	<u>Response Time</u>
Low	High
High	Low

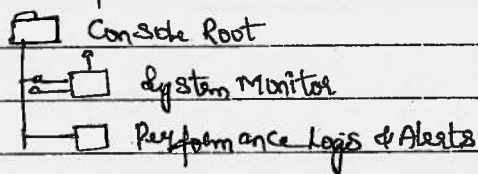
(b) Throughput: Number of transactions executed by db system for a period of time (1 min).

<u>Performance</u>	<u>Throughput</u>
High	High
Low	Low

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'Performance' window is displayed.



Color	Scale	Counter	Instance	Parent	Object	Computer
	1.000	Pages/sec	...	...	Memory	11 Peers
	100	Avg disk queue length	-Total	...	Physical Disk	11 Peers
	1.000	% Processor time	-Total	...	Processor	11 Peers

Click + button

In 'Add Counters' window

- ☐ Use Local Computer Counters
- ☒ Select Counters from Computer:

11 Peers ▼

Performance Object:

Memory ▼

• Processor

• 'SQL Server': Buffer Manager

• 'SQL Server': Data Bases

☐ All Counters

☒ Select Counters from list

Add

Explois

☐ All instances

☒ Select instances from list.

Close

Note: click [Icon] button

11 Peers

→ Memory

Committed Bytes

Pages/sec

→ Processor

% Processor time

SQL Server

DATE

Sample data Every:

Interval: Units: * Now Select **Log files** tab

Log file Type and Name:

Log file Type

 Configure☒ Textfile (Comma delimited)☒ End file names with: Start numbering at: Example: D:\Perf logs\demoSql Counters - ^{months}02^{date}11^{hours}09.csvComment: * Now Select **Schedule**

Start log:

☐ Manually (using the Short Cut Method.) ^{Menu}☒ At on → ^{Format} month/date/year

Stop log:

☐ Manually (using the Short Cut Menu)☐ After Units: ☐ At on ☐ when the log file is full.

when a log file closes:

☐ Start a new log file.☐ Run this Command Browse

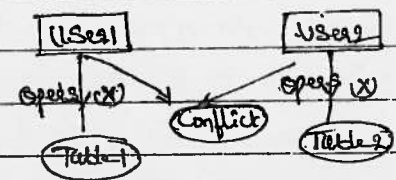
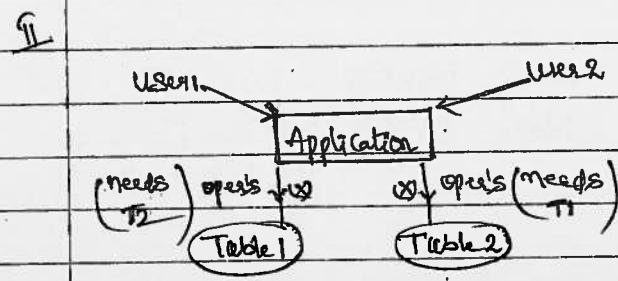
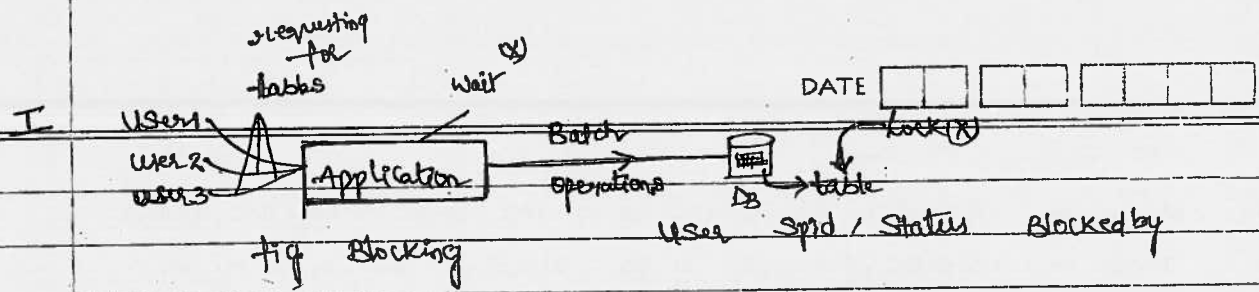
Note: Select Required options given above and

click **Apply**click **OK**

Note: In 'Counter logs' - a 'Counter log' file is created.

Note: To Read performance Objects & Counters (of Sql Server only),
we use 1 DmV.

Select * From Sys.dm-OS-Performance-Counters.



Eg: Blocking Situation

Connect to a particular Sql Server Instance (Eg: Peers)

-- Connection 1: SqlUser1

-- Connection 2: SqlUser1

Use Testdb

Use Testdb

Go

Go

(X) lock

Begin Transaction

Begin Transaction

Delete from Emp

Select * from Emp

-- Press F5

-- Press F5.

* To Trace the Blocking

→ Connect to Sql Server Instance (Eg: Peers) as an Administrator.

View the users processes using the following T-Sql Code:

T-Sql Code

Select * from sysprocesses
Where Spid > 50 and blocked <> 0

* To Verify the Sql Statements Submitted by the user processes.

Eg: DBCC Inputbuffer (Spid)
DBCC Inputbuffer (51)

* To Verify / Know the locks applied internally by the processes.

Ex: Exec Sp_Lock 51.

* To Verify the database name

Go Select db_name(11)

Use Testdb

Select Objectname (XXX)

Select Object-name (XXX)

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Database Level Tuning :

To perform db level tuning, we have to consider the following.

- Database Space Monitoring & Shrinking database.
- Indexes Fragmentation
- Statistics Maintenance
- Table Level Maintenance

Space growth
indexes
statistics
data partitions.

I(a) DATABASE SPACE MONITORING & SHRINKING DATABASE

* To verify the space used by the database

Syntax { use dbname
 free sp - space used

* To verify log-file space usage of all databases.

Syntax : dbcc sqlperf (logspace)

* To verify a particular db files related information.

use dbname

Select * from sysfiles

-- in the result, verify/check 'Unallocated' column

-- if this column value is close to 'zero' then db space is full

dbcc sqlperf (log space)

-- (Logspace used) % column when closes to 100%
means log file have no free space.

Q. Database has no free space?

- Solns
- Shrink the db or datafiles
 - Alter database Modify Datafile
 - Alter database Add Datafile.

(informing)

Shrinking : The process of returning free space to sql server.

we can shrink the complete db (or) each datafile / log file / file group.

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Log file is full :-

- ① Shrink the log file
- ② Perform Transaction log Backup with 'Truncate-only'.
Backup log dbname
TO disk = 'filepath'
with 'Truncate-only'.
- ③ Change the db state
 - * User Access mode = single
 - * Recovery Mode = Simple
 - * Stop and Start the db → log space becomes zero
 - * Recovery Model to : full
 - * User Access mode = Multi-user
- ④ Alter Database Modify Log file
- ⑤ Alter Database Add log file

II

INDEX FRAGMENTATION :

Index is an ordered list of values taken from one or more columns of a table and organized into ^{Balanced} B-Tree Structure.

Usage : Indexes will improve the performance of 'Select' queries.

Types of Indexes

- ① clustered Indexes
- ② non-clustered Indexes
- ③ Full-Text Indexes ← 'String' datatype
- ④ XML indexes (New in SQL Server 2005)

- * Index can be created on a single column of a table / multiple columns of a table.
- * Only one clustered index creation is possible on a table.
- * By default, 'clustered index' is created on primary key column of a table.
- * A table can have 249 non-clustered indexes.
- * By default, 'non-clustered index' is created on 'unique' constraint column.
- * clustered & non-clustered indexes can be created on any datatype column.
- * Full Text indexes can be created on only 'String' datatype column.
Eg: char, varchar, text, nchar, nvarchar, ntext.
- * XML indexes can be created on only XML datatype column.

S - System tables
 U - User tables
 V - views
 P - stored procedures

FN - User's
 trig - triggers
 K - Key Columns
 R - Rules
 D - default

(96)

use dbname
 Go
 Select object-name(id)
 from Sysindexes

-- To list out all objects in a db

Use dbname
 Go Select * from Sysobjects

where type = 'S' = 'V' = 'PK'
 = 'U' = 'P' ;
 = 'K' = 'SO'

Eg: Use Testdb
 Go

Select * from Sysobjects

Assignment:

- * List all tables in a db not having atleast one index } Use Sysindexes & Sysobjects tables and apply join.
- * List all tables in a db having atleast one index

~~Rebuild~~ ~~Reorganize~~ ~~Rebuild~~ ~~Reorganize~~ ~~Rebuild~~ ~~Reorganize~~
 Rebuild Indexes (clustered) → insert short poor performances → drops & re-create the B-tree structure with specified fill factor
 → Periodically
 → Fill factor — leaf pages
 → B-tree is re-created
 → Takes more time because dropped & re-created.
 → Pad index — fill factor is applied to intermediate pages & leaf pages.

~~Rebuild~~ ~~Reorganize~~ ~~Rebuild~~ ~~Reorganize~~
 Reorganize Indexes → No fill factor
 → drops the leaf pages only.
 → Non-clustered.
 → ~~Reorganize~~ ~~Reorganize~~ ~~Reorganize~~
 → Rearrange the leaf pages in the B-Tree

-- Syntax: (table level) → means all indexes of the table.

Alter index <indexname/all>

on Tablename

Reorganize | Rebuild with (fill factor = ?)

-- Eg: Alter index Salindex
 on Emp

Rebuild with (fill factor = 75)

} for single table.

Fill Factor: Means how full is each leaf-page in the B-structure
 Recommended value — (70% to 80%.)

Pad index: Fill factor % is applicable — to all intermediate pages, root pages in the B-Tree also

Query optimizer → Execution plan

↳ Cost of Execution plan

↳ which indexes to be used

↳ index Stats — more ✓ — Selected

— less x — not Selected.

III STATISTICS MAINTENANCE :

- * For every index — Statistics is maintained automatically. Column
- * 'Statistics' means selecting / selectivity or density of an indexed Column / non c.i.n —
- * Sql Server maintains Statistics for indexed Columns automatically.
- * We have to create and maintain Statistics for non-indexed Columns — Explicitly (Need to be updated periodically).
- * Explicitly we can create 'Statistics' for Combination of 2 or more Columns also.

Query optimizer

- * Sql Server Query optimizer is Cost based optimizer.
- * This will prepare 'Execution plans' for queries & stored procedures by considering 3 things.

(a) indexed Columns Statistics

(b) Non-indexed Column Statistics.

(c) Cost of the Execution plan (cpu cost, I/O cost)

Note: default Statistics — single Columns

Combine Statistics — Multiple Columns

Explicit Statistics — Need to be updated periodically.

Indexed Columns — updated automatically.

Non-indexed Columns — we have to create Statistics.

Implicit Statistics — for single Columns.

Explicit Statistics — for Combination of Columns.

Note: By default, ^{name} Statistics of an indexed Column (~~non-indexed~~) is
index name.

- * To view the 'Statistics' of an indexed Column / non-indexed Columns :

System DBCC Show-Statistics (Table name, Stat name)
(or) index name

Eg: Use Testdb

Go

DBCC Show-Statistics ('Emp', 'Salindex')

index name of 'Sal' Column

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**
**
*

Practical : How to create a Maintenance plan for shrinking db, indexes - maintenance, statistics Maintenance and Consistency checking for or more db's?
 -- Creating a Maintenance plan

Solution : Open SSMS - SQL Server management Studio.

In Object Explorer

↳ Connect to a required 'SQL Server instance' (Eg: Peers)

R/C - Right click

↳ Expand "Management"

R/C on 'Maintenance plan' and

Choose 'Maintenance plan wizard'

In 'New Maintenance plan wizard' page

click **Next**

*
**

In 'Select a Target Server' Page

Name : **Database Tuning plan**

Description : **All databases mainten. plan**

Server : **Peers** ... → Click here to Connect to a particular server

☒ Use Windows Authentication

☐ Use SQL Server Authentication

User Name :

Password :

Help

Back

Next

Finish

Cancel

click **Next**

*
**

In 'Select Maintenance Tasks' Page

Select one or more Maintenance Tasks :

☒ Check database integrity

☒ Shrink database

☒ Reorganize index

☒ Rebuild index

☒ Update Statistics

☒ Clean up History

☒ Execute SQL Server Agent Job

☐ Backup Database (Full)

☐ Backup Database (Differential)

☐ Backup Database (Transaction Log)

click **Next**

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*
** In 'Define Reorganize Index Task' page

Databases :

All databases	▼
---------------	---

object :

	▼
--	---

Selection :

	▼
--	---

☒ Compact large objects

click

Next

*
** In 'Define Rebuild Index Task' page

Databases :

All databases	▼
---------------	---

object :

	▼
--	---

Selection :

	▼
--	---

Free Space Options :

☐ Reorganize pages with the default amount of free space.

☒ change free space per page percentage to

25

 %.

Advance options :

☒ pad index

☒ Sort results in tempdb

☐ ignore duplicate keys

☒ Keep index online while reindexing

click

Next

*
** In 'Define update Statistics Task' page

Databases :

All databases	▼
---------------	---

object :

	▼
--	---

Selection :

	▼
--	---

update :

☒ All Existing Statistics.

☐ Column Statistics only

☐ Indexes Statistics only

click

Next

*
** In 'Define cleanup History Task' page

Select the historical data to delete :

☒ Backup & Restore History.

☒ SQL Server Agent Job History.

☒ Maintenance plan History.

Remove historical data older than :

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4	▲▼
---	----

Weeks	▼
-------	---

click

Next

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*
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In 'Select Report options' page.

Note: Click here to select a path

☒ Write a Report to a text file:

Folder Location: D:\program files\Microsoft SQL Server\MSSQL.1\MSSQL\LOG\...

☒ Email Report

TO TestAdmin v

Click Next

*
**

In 'Complete the wizard' page

Verify the choices made in the wizard, then click Finish.

Click Finish to perform the following actions:

=====

Click Finish

*
**

In 'Maintenance plan wizard progress' page

click stop to interrupt the operation

☒ Success ← Status

5 total

0 Error

Details:

5 success

0 warning

Action	Status	Message
• Creating Maintenance plan 'DB mp1'	Success	
• Adding Tasks to the mp	✓	
• Adding Scheduling options	✓	
• Adding Reporting options	✓	
• Saving M.P. - DB mp1	✓	

Stop

Report v

↓ View Report

Save Report to file

Copy Report to clipboard

Send Report as Email.

Finally, click Close

Verification: In object Explorer

[-] [S] Server (SQLServer 9.0.1399 - Server Administrator)

Expand 'Management'

Expand 'Maintenance plans'

Here a new M.p just created 'DB mp1' will appear in the list.

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In 'Trace Properties' window

Select **General** tab. **General** **Events Selection**Trace Name : Trace provider Name : Trace provider Type : Version : → Use the Template : ☒→ ☒ Save to file : ☒

In 'Save As' window

File Name : Save as type : Set Max file size (MB) : ← default size☒ Enable file rollover☐ Server processes trace data→ ☐ Save to Table : ☒☐ Set Minimum Rows (in thousands) → ☐ Enable Trace stop time : ☒ ☒

Now, Select 'Events Selection' tab

check ☒ Show all events.

Events	Textdata	Duration	SPID	EBID	DBName	Object type	Login Name
☒ TSQL							
☒ SqlBatch Completed	☒	☒	☒	☒	☒		☒
☒ SqlBatch Starting	☒		☒	☒	☒		☒
☒ Stored procedures							
☐ Rpc: Completed	☒	☒	☒	☒	☒		☒
☐ Rpc: Starting	☒		☒	☒	☒		☒

Now, click on **Column filters**

In 'Edit filter' window

DatabaseId

☒ Database Name☒ like

Duration

└─

Login Name

Click

Object type

SPID

TextData
classmateFinally, click **RUN**
'profiles' will be started and records/captures all queries/spid

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□ Enable Trace Stop Time: 2/12/2011 8:07:38 Am

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Syntax: ① Select * From Table
 with (index = indexname) (index hint)
 where <Condition>

② Select * From Table (Tablockx) }
 ③ Select * From Table (NoLock) } locking hints.

(5) TO optimize : ReCompile the Stored Procedure.

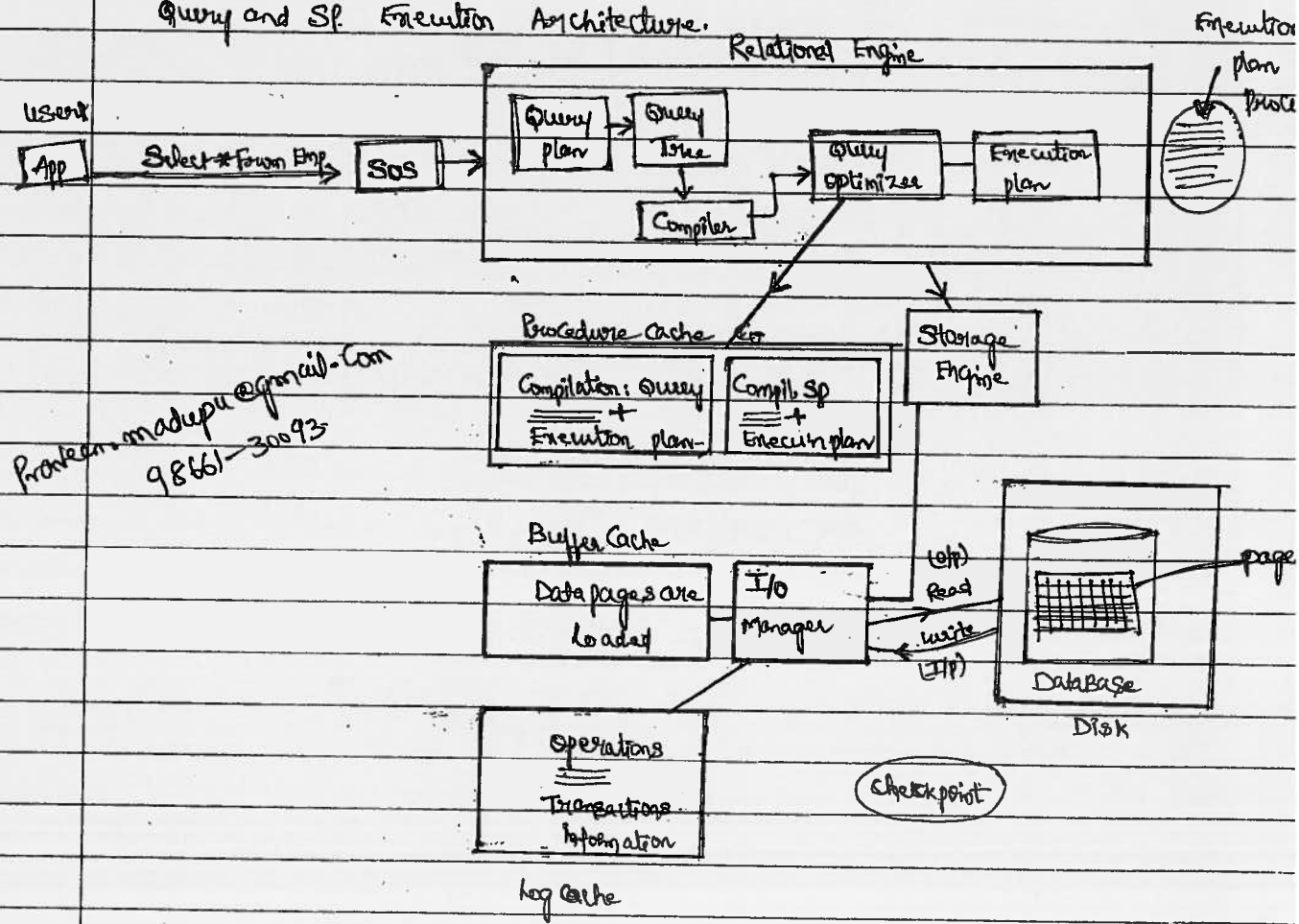
Exec Sp-ReCompile < procedure Name >

* Once we ReCompile the 'SP', it will remove (SP Compiled version) + 'Execution plan' from the 'Procedure Cache'.

* This will ReCompile the (Procedure) and 'Execution plan' prepared everytime when a user calls the SP.

Note: Generally, 'ReCompile' option is recommended for the SP's that are to be executed occasionally. (Weekly once / Monthly once) for maintenance - purpose.

Query and SP Execution Architecture.



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